

A phase II study assessing safety and efficacy of Cabozantinib for advanced or metastatic cervical carcinoma after platinum treatment failure

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REGLEMENTARY CLASSIFICATION OF TRIAL: RIPH 1

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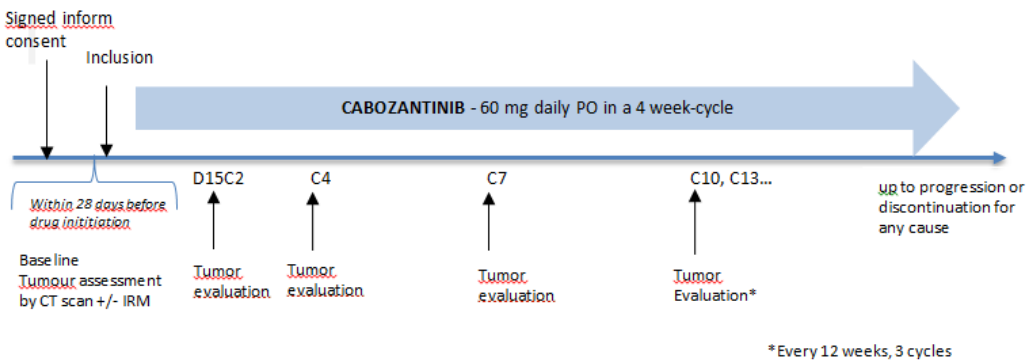
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1 SYNOPSIS

TITLE	Phase II study assessing safety and efficacy of Cabozantinib for advanced or metastatic cervical carcinoma after platinum treatment failure.
ACRONYM	CABOCOL-01
Coordonnator	Dr Elodie COQUAN, medical oncologist, Centre François Baclesse, Caen
Indication	Recurrent unresectable or metastatic cervix carcinoma
Design	A phase II single-arm two-stage multicentre study
Objectives	Main objective To assess efficacy and safety of cabozantinib in monotherapy in advanced/metastatic cervical cancer (CC) after failure of platinum-based regimen treatment.
	Secondary objectives To assess - The safety profile of cabozantinib - The objective response (RECIST v1.1 criteria) - The progression-free and overall survival - The health-related quality-of-life (QoL) of patients - The nutritional outcomes To constitute biological collections (blood, tumoral tissue) for further biological explorations such as: - Evaluation of the impact of immune and angiogenesis biomarkers and intra-cellular activated pathways (blood, ctDNA and tissue) in terms of oncological outcomes and response to treatment, - Exploration of the modulatory role of cabozantinib on immune cells, - Evaluation of the predictive value of HPV-ctDNA during cabozantinib treatment
Judgement criteria	Main criterion The main objective will be based on joint primary endpoints of efficacy and safety as proposed by the Bryant-and-Day design: • Efficacy assessed by the proportion of patients with disease control rate 3 months after cabozantinib treatment initiation. • Safety assessed by the proportion of patients with clinical gastro-intestinal (GI) perforation/fistula, GI-vaginal fistula and genito-urinary (GU) fistula events grade ≥ 2 (NCI CTCAE v 5.0) occurring up to 1 month after the end of treatment
	Secondary criteria The secondary endpoints are : ○ Progression-free survival, defined as the time between initiation of cabozantinib treatment and progression (RECIST 1.1 criteria) or death of any cause whichever occurs first. ○ Objective response rate defined as the percentage of patients who have achieved complete response or partial response with RECIST 1.1 criteria. ○ Overall survival, defined as the time between initiation of cabozantinib treatment and death of whatever cause. ○ Toxicities evaluated according to NCI CTCAE v5.0 criteria, in terms of kind, grade, time of onset, reversibility ○ Significant toxicities responsible for interruption or dose reduction of cabozantinib treatment ○ Proportion of patients requiring cabozantinib dose reduction or treatment stopping for toxicity ○ Scores of Quality-of-life according to French versions of the self-administered standardized validated questionnaire: EORTC QLQ-C30 and its additional cervix cancer module (QLQ-CX24) ○ To investigate nutritional outcomes, including: weight, sarcopenia, biological markers, muscle mass, and muscle strength under cabozantinib in advanced cervical carcinoma

Inclusion criteria	- Female 18 years of age or older - Histologically confirmed recurrent unresectable or metastatic cervix carcinoma with squamous cell, adenocarcinoma or adenosquamous histology - Patient may have received at least one prior chemotherapy regimen of platinum-based chemotherapy for recurrence or metastatic disease. Cisplatin given in combination with radiation for a localized disease does not count as a prior chemotherapy. Prior treatment for advanced/metastatic disease with bevacizumab is allowed. Prior treatments with immune checkpoint inhibitors are allowed. - ECOG performance status 0-2 - Measurable disease per RECIST 1.1 - The subject must have recovered to baseline or CTCAE v.5.0 (Common Terminology Criteria for Adverse Events, version 5.0) ≤ Grade 1 from clinical toxicities related to any prior treatments, i.e chemotherapy or pelvis radiation unless AE(s) are clinically non-significant (for example alopecia) - Adequate organ and marrow function, defined as follows, based upon laboratory tests performed within 7 days before inclusion: - o Absolute neutrophil count (ANC) ≥ 1000/mm³ (≥ 1.0 GI/L) - o Platelets ≥ 100,000/mm³ (≥ 100 GI/L) - o Hemoglobin ≥ 10 g/dL (≥ 100 g/L) (red blood cell transfusion is allowed) - o Total bilirubin ≤ 1.5 fold the upper limit of normal (for subjects with Gilbert's disease, ≤ 3 mg/dL or ≤ 51.3 μmol/L) - o Serum albumin ≥ 3.0 g/dL (≥ 30 g/L) - o Calculated creatinine clearance ≥ 30 mL/min by the CKD-EPI method. - o Alanine aminotransferase (ALT) and aspartate aminotransferase (AST) < 3.0 x the upper limit of normal - o Urine protein/creatinine ratio (UPCR) ≤ 1g/g (≤ 113.17 mg/mmol creatinine) or 24-hour urine protein < 1 g - Left-ventricular ejection fraction ≥ 50% - Subjects affiliated to an appropriate social security system - Female subjects of childbearing potential must not be pregnant at screening and during treatment by Cabozantinib. Effective methods of contraception should be used throughout the course of treatment and for at least 4 months after the end of treatment. Sexually active fertile subjects and their partners must agree to use medically accepted barrier methods of contraception (e.g., male or female condom) during the study and 4 months after the last dose of study treatment, even if oral contraceptives are also used. - Before patient registration, written informed consent must be given according to ICH/GCP, and national/local regulations.
Non inclusion criteria	- Clinically significant gastrointestinal abnormalities that may increase the risk for gastrointestinal bleeding and/or fistula / perforation including, but not limited to: Active peptic ulcer disease, inflammatory bowel disease (e.g. ulcerative colitis, Crohn's disease), history of abdomino and/or pelvic fistula, gastrointestinal perforation, or intra-abdominal abscess, gastro-intestinal obstruction - Patients with lesions on baseline pelvic MRI which may major the risk of abdominal and/or pelvic fistula/perforation - Clinically significant gastrointestinal abnormalities that may affect absorption of investigational product including, but not limited to: malabsorption syndrome, major resection of the stomach or small bowel. - Previously identified allergy or hypersensitivity to components of the study treatment formulations (Note: patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take cabozantinib and are also excluded).

	- History of any one or more of the following cardiovascular conditions within the past 6 months: Cardiac angioplasty or stenting, myocardial infarction, unstable angina, coronary artery bypass surgery, symptomatic peripheral vascular disease, class III or IV congestive heart failure, as defined by the New York Heart Association (NYHA), serious cardiac arrhythmias. - Corrected QT interval (QTc) calculated by the Fridericia formula > 500 msec within 28 days before inclusion (see Annex for Fridericia formula). Note: if initial QTcF is found to be > 500 msec, two additional ECGs separated by at least 3 minutes should be performed. If the average of these three consecutive results for QTcF is ≤ 500 msec, the subject meets eligibility in this regard. - Uncontrolled hypertension defined as systolic blood pressure (SBP) of > 150 mmHg or diastolic blood pressure (DBP) of > 100 mmHg despite an optimal treatment. - History of cerebrovascular accident including transient ischemic attack (TIA), symptomatic pulmonary embolism or untreated deep venous thrombosis (DVT) within the past 6 months. <i>Note: Subjects with recent DVT or asymptomatic pulmonary embolism who have been treated with therapeutic anti-coagulating agents for at least 4 weeks are eligible.</i> - Major surgery or trauma within 28 days prior to first dose of investigational product and/or presence of any non-healing wound, fracture, or ulcer. - Evidence of active bleeding or pathologic conditions that carry high risk of bleeding such as coagulopathy or tumor involving major vessels. - At least 6 weeks must have elapsed between the last dose of pelvis palliative radiation and the first dose of cabozantinib or 2 weeks for other localization of palliative radiation - Presence of brain metastases or epidural disease unless adequately treated with radiotherapy and/or surgery (including radiosurgery) and stable for at least 3 months before inclusion. Eligible subjects must be neurologically asymptomatic and without corticosteroid treatment at the time of inclusion. - Concomitant use of known strong CYP3A4 inhibitors or inducers. - Patients with second primary cancer, except adequately treated non-melanoma skin cancer, or other solid tumors curatively treated with no evidence of disease for ≥ 3 years - Any psychological, familial, sociological or geographical condition potentially hampering compliance with the study protocol and follow-up schedule; those conditions should be discussed with the patient before registration in the trial - Concurrent participation in any therapeutic clinical trial - Patient deprived of liberty or placed under the authority of a tutor
Experimental Plan	Eligible patients who have completed screening and have signed the informed consent to participate to this two-step phase II trial will receive Cabozantinib treatment.  Cabozantinib will be administered at the daily dose of 60 mg given orally in a 4-week cycle. It will be continued without interruption until disease progression or discontinuation for any cause.

Number of patients	The total sample size to be enrolled is 57 patients. As it is the first time the cabozantinib is used for advanced cervical carcinoma after platinum treatment failure, we plan to use a single arm two-stage multicentre phase II based on a Bryant-and-Day design. This single arm two-stage design was selected in order to simultaneously assess efficacy and safety and minimize the expected number of patients treated in the event that cabozantinib monotherapy proves to be very disappointing (in terms of efficacy and/or safety) in this setting. In accordance with the literature estimating a median PFS around 3.4 months for advanced cervical cancer patients under bevacizumab monotherapy and 4 months for pazopanib. By extrapolation, we expect for: Efficacy: $\pi_{\text{Efficacy}0} = 30\%$, $\pi_{\text{Efficacy}1} = 50\%$, the unacceptable and acceptable 3-month control rates Toxicity: $\pi_{\text{Toxicity}0} = 25\%$, $\pi_{\text{Toxicity}1} = 10\%$, the unacceptable and acceptable fistula/perforation rates Assuming the error rates: - One-sided $\alpha_{\text{Efficacy}}=5\%$ and $\alpha_{\text{Toxicity}}=10\%$ - Power=80% <u>Stage 1:</u> 22 assessable patients will be included for <u>interim analysis</u>. Initially, enrolment should not be halted to conduct the interim analysis. However, due to the quickness of recruitment in the study, the protocol is amended to suspend the inclusions during the interim analysis, in order to have a sufficient perspective on the tolerance and efficacy of the treatment. Then: ⇒ if there are $p_{\text{Efficacy}} < 8/22$ OR $p_{\text{Toxicity}} > 4/22$ the study will STOP. ⇒ Otherwise the second stage (<u>stage 2</u>) will be planned with 29 additional assessable patients (for a total of N=51 patients). At the final analysis: If $p_{\text{Efficacy}} < 21/51$ OR $p_{\text{Toxicity}} > 9/51$: the study will be considered as negative. Assuming a 10% drop-out rate, a total of 57 (25 for the stage 1 and 32 additional for the stage 2) patients will be included in the present study. Among 57 patients : 19 evaluable bevacizumab pre-treated patients Using a 1-stage A'hern design (Reference: A'Hern RP. Sample size tables for exact single-stage phase II designs. Stat Med. 2001 Mar 30;20(6):859-66.) in the bevacizumab pre-treated cohort, would allow to accept 3-month control rates $\geq 50\%$ while rejecting a rate of $\leq 20\%$ with 80% power at a 2.5% significance level. The resulting decision rule requires to observe ≥ 8 “alive-controlled” (responding to the treatment) patients at 3 months out of 19 evaluable patients in the bevacizumab pre-treated cohort to reject the null hypothesis ($\leq 20\%$) in favor of the alternative ($\geq 50\%$).
Centres	Number of participating centres : about 8
Study duration	• Start of study: 1th trimester 2020 • Total inclusion period : 30 months • Total study period : 42 months (30 months and about 1 year for follow-up)

2 STUDY FLOW CHART

	Before inclusion within 28 days before drug initiation	During treatment (1 cycle = 28 days)					End of treatment 30 days after the last dose of study treatment (+/- 7 days)	Follow-up every 3 months up to progression	Overall survival after disease progression
		Cycle 1		Cycle 2		Other Cycles from cycle 2			
		D1	D15	D1	D15	D1			
Study drug administration		CABOZANTINIB treatment (in a 28 day-cycle)							
Informed Consent <i>should be done before any study procedures</i>	✓								
Clinical assessment - Complete physical examination including gynecological examination, weight, height (only at baseline), ECOG, vital signs - Adverse Events collection and concomitant treatments	✓	✓	✓	✓	✓	✓	✓	✓	
Laboratory Assessments¹ - Blood assessment - Urinary assessment	✓ within 14 days before drug initiation	✓ ³	✓ within 3 days	✓ within 3 days	✓ within 3 days	✓ within 3 days	✓ within 3 days	✓ within 3 days	✓ within 3 days
Cardiac assessment - ECG (QT interval) - Cardiac Echography or MUGA	✓ ✓			✓			✓		
Quality-of-life assessment EORTC QLQ-C30 / CX24	✓				✓		✓		
Diary card		✓ ⁴		✓		✓			
Radiological assessment - CT-scan (thorax, abdominal and pelvis) ⁶ - Pelvic MRI ²	✓ ✓				within 3 days ✓ ✓		+/- 7 days ✓ ✓		+/- 14 days ✓ ✓
Biological collections - Tumoral biopsy <u>optional</u> ⁵ - Blood samples	✓ ✓			✓	✓		✓ at progression		

¹ Laboratory assessment

- Hematology (CBC, platelets)
- Serum biochemistry (Albumin, total alkaline phosphatase (ALP), alanine amino transferase (ALT), aspartate amino transferase (AST), corrected calcium, creatinine and clearance by CKD-EPI method, glutamyltransferase (GGT), glucose, lactate dehydrogenase (LDH), magnesium, phosphorus, potassium, sodium, total bilirubin (conjugated and unconjugated if clinically indicated), total protein
- Coagulation (PT/INR and PTT) mandatory before inclusion, then for other visits only if clinically indicated
- Thyroid function tests (TSH, free T3, free T4) (at each biological assessment except D15C1 and D15C2)
- Tumor marker : SCC antigen (at each biological assessment except D15C1)
- Urine analysis :
 - Before inclusion: Urine analysis: urine protein-to-creatinine ratio (UPCR) ≤ 1 g/g (≤ 113.2 mg/mmol) creatinine or 24-hour urine protein < 1 g. When a UPCR exceeds 1g/g, a repeat UPCR or a 24-hour urine protein and creatinine should be performed to confirm the result
 - For other visits: Urine dipstick (with protein, blood and leucocytes).
If results > 2+, an urinary analysis must be performed with a Urine protein / creatinine ratio (UPCR). When a UPCR exceeds 1 g/g, a repeat UPCR or a 24-hour urine protein and creatinine should be performed to confirm the result.
- Urinal pregnancy test (Women of childbearing potential), only before inclusion

² Mandatory pelvis IRM at inclusion and optional for further evaluations. MRI can be used in addition of CT scan if a local recurrence could not be assessed by CT

³ Only if realized more 14 days before D1

⁴ Subjects should be instructed before the beginning of the treatment of the risk of diarrhea and the initial management. A preventive prescription can be given to the patient at the beginning of the treatment.

⁵ Fresh biopsy from primitive and/or from metastatic sites if feasible and only if the patient agrees on the consent form.
AND Mandatory: One paraffin block of archival initial or recurrent tumour will be sent to sponsor during study.

⁶ A central review of the scanners will be done in the study to assess sarcopenia. An anonymized copy of the scanner imagery will be sent to the sponsor during the study.

3 SCIENTIFIC RATIONALE OF THE STUDY

3.1 EPIDEMIOLOGY AND CURRENT MANAGEMENT OF ADVANCED CERVICAL CANCER

Cervical cancer is the second most commonly diagnosed cancer and the third leading cause of cancer death among females in less developed countries (Torre et al. 2012). The most significant cause of cervical cancer is persistent human papillomavirus infection (HPV). HPV is detected in 99% of cervical tumors, particularly the oncogenic subtypes such as HPV 16 and 18. Widespread HPV vaccination and cytologic screening have enabled a decline in the incidence of cervical cancer (McGraw et al. 2014). Early-stage and locally recurrent disease may be cured with radical surgery, chemo-radiotherapy or both. However, if disease persists or recurs, options are limited and the prognosis is poor.

❖ Role of antiangiogenic treatments

Palliative chemotherapy with cisplatin 50 mg/m² every 3 weeks was the standard of care of advanced/metastatic CC, until cisplatin-based doublets with topotecan or paclitaxel demonstrated their superiority in terms of response rate (RR) and progression-free survival (PFS): objective responses occurred in 36% receiving cisplatin and paclitaxel (versus 19% for cisplatin alone) (Long et al. 2005; Moore et al. 2004). The median PFS was 4.8 months for the combination but there was no difference in median overall survival (OS) (9.7 months).

Vascular endothelial growth factor (VEGF) promotes angiogenesis and is an important mediator of disease progression in cervical cancer. Bevacizumab prevents tumour angiogenesis by blocking vascular endothelial growth factor and was shown to be active in a phase II study (GOG-227C) in recurrent cervical cancer (Monk et al. 2009). Based on this observation, the GOG-240 study explored the addition of bevacizumab to chemotherapy in a randomized phase III trial with a 2x2 factorial design in which OS was the primary end-point (Tewari et al. 2014). Patients with primary stage IVB or recurrent/persistent disease, good PS (0 or 1) and measurable disease were randomized to paclitaxel–cisplatin or paclitaxel–topotecan, both with or without bevacizumab. With the data for the two chemotherapy regimens combined, the addition of bevacizumab to chemotherapy was associated with increased OS (17.0 months vs. 13.3 months; hazard ratio for death, 0.71; 98% confidence interval, 0.54 to 0.95; P = 0.004) and higher RR (48% vs. 36%, P = 0.008). Bevacizumab, as compared with chemotherapy alone, was associated with an increased incidence of hypertension of grade 2 or higher (25% vs. 2%), thromboembolic events of grade 3 or higher (8% vs. 1%), and gastrointestinal fistulas of grade 3 or higher (3% vs. 0%). But, there was no deterioration in health-related quality of life (Penson et al. 2015).

So, bevacizumab associated with a doublet of platinum chemotherapy is the standard of care of the first line regimen in metastatic or recurrent cervical cancer which is not eligible to local treatment.

In case of bevacizumab treatment or, more generally of antiangiogenic treatment, increased reports of fistulas have been reported in previous studies. In the phase III trial assessing efficacy of bevacizumab combined with chemotherapy vs chemotherapy alone in first line treatment of metastatic cervical carcinoma (Tewari et al. 2017), 32 (15%) of 220 patients in the chemotherapy plus bevacizumab group had fistulas all of whom had had previous radiotherapy, compared with three (1%) in the chemotherapy-alone group, also all of whom had had previous radiotherapy. 13 (6%) patients had clinically significant or severe (ie, grade 3) fistula in the chemotherapy plus bevacizumab group versus one (<1%) in the chemotherapy-alone group. No fistulas resulted in surgical emergencies, sepsis, or death, and in addition to pelvic irradiation, other factors associated with fistulas included pelvic disease, pre-existing hypertension, and current tobacco use. In their real life data study, Godoy-Ortiz et al. report increase rate of fistula (22%) over 27 patients (Godoy-Ortiz et al., 2018). Palavalli studied predictive factor of fistula event in 74 patients treated for advanced, recurrent or metastatic cervical carcinoma (Palavalli Parsons et al., 2017). Lower albumin levels and use of bevacizumab were identified as an independent predictor factors for fistula (P=0.004 and 0.024, respectively).

Other different additional agents that target VEGF have been investigated in advanced cervical cancer such as pazopanib, an intracellular small molecule tyrosine kinase inhibitor (TKIs) receptor, sunitinib and cediranib (Symonds et al. 2015).

- In 2010, Monk enrolled a total of 410 patients to assess the efficacy of pazopanib, lapatinib or the combination of both, in women with metastatic, persistent or recurrent cervical cancer. Unfortunately, the study prematurely discontinued for futility and excessive toxicity of the combination, although patients who received pazopanib in monotherapy experienced a higher PFS and longer median OS (12.4 months) when compared patients treated with lapatinib. Diarrhea was the most common G3 adverse event: 11% for pazopanib and 13% for lapatinib (Monk et al. 2010). Because only 50% of the total expected patients participated in the final analysis, it was not possible to draw any definitive conclusions regarding the real effectiveness of the therapy.
- Cediranib is an inhibitor of the tyrosine kinase activity of VEGFR1, 2, and 3, and c-KIT. A randomized phase II study evaluated the efficacy of cediranib associated with carboplatin and paclitaxel in the first line treatment of metastatic/advanced cervical cancer. The trial closed prematurely on July 27, 2012, owing to withdrawal of drug supply but results showed a significant efficacy of the combination (median PFS 8.1 months than in the placebo group 6.7 months with a hazard ratio of 0.58, $p = 0.032$). This finding was accompanied by an increase in toxic effects, mainly diarrhea, hypertension, and febrile neutropenia.
- In a phase II study, sunitinib was administered to 19 patients with unresectable, locally advanced or metastatic cervical cancer (Mackay et al. 2010). Sixteen patients (84%) had a stable disease as their best response with a median duration of 4.4 months. Five patients (26.3%), who had previously received either chemo-radiation or radiation therapy, developed a fistula during or after sunitinib assumption. It was therefore concluded that the drug has insufficient activity and was too toxic in this clinical setting.

Following progression after first-line chemotherapy, different cytostatic agents such as docetaxel, topotecan or gemcitabine, have been evaluated (Garcia et al. 2007; Muderspach et al. 2001; Schilder et al. 2005). However, response rates are low and duration of response is short.

Very recently, pembrolizumab was investigated in 98 patients with recurrent or metastatic cervical cancer who had received at least one line of chemotherapy for metastatic disease in a cohort of KEYNOTE 158 (NCT02628067), a multicenter, non-randomized, open-label, multi-cohort trial (Hyun et al. 2018). Patients were treated with pembrolizumab until unacceptable toxicity or documented disease progression. Among the 98 patients, 77 (79%) patients had tumors that expressed PD-L1 with a combined positive score (CPS) ≥ 1 . With a median follow-up time of 11.7 months, the objective response rate (ORR) in 77 patients was 14.3% (95% CI: 7.4, 24.1), including 2.6% complete responses and 11.7% partial responses. The estimated median response duration based on 11 patients with a response was not reached; 91% had a ≥ 6 months response duration. No response was observed in patients without tumoral PD-L1 expression (CPS < 1). Based on these results, the Food and Drug Administration approved pembrolizumab, in June 2018, for recurrent or metastatic cervical cancer patients with disease progression or after chemotherapy, whose tumor expresses PD-L1 (CPS ≥ 1).

However, in Europe, to date, no standard second-line chemotherapy is recommended in guidelines (Marth et al. 2017).

❖ Two interesting targets in cervix carcinoma: AXL and MET

HPVs are small, non-enveloped viruses that induce squamous epithelial tumours. HPV16 and 18 are the most prevalent types and are associated with genomic integration and cervical cancer.

Axl receptor tyrosine kinase is involved in the tumorigenesis and metastasis of many cancers. In a recent report, Axl expression was markedly higher in HPV type 16E6 (HPV16E6)-overexpressing HeLa (HE6F) cells than in the controls. Axl expression was regulated by HPV16E6-mediated PTEN/AKT signaling pathway, and Axl promoter activity was regulated through myeloid zinc finger protein 1 (MZF1) activation in cervical cancer, which promoted malignancy. The expression of Axl and MZF1 was highly correlated with clinical stage of cervical cancer and HPV16/18 infection. Taken together, Axl expression was induced by HPV16E6 in cervical cancer cells, suggesting that blockade of Axl signaling might be an effective way to reduce the progression of cervical cancer (Lee et al. 2017).

Similarly, c-MET receptor and its ligand, the hepatocyte growth factor (HGF) seems to be an interesting target in cervical carcinoma. The c-MET oncogene is known to play an important role in cancer growth and metastasis as well as development of drug resistance, especially with VEGFR inhibitors. In cervical cancer, c-MET and HGF overexpression tend to have a prognostic value. Zhang and al found that patients with cervical squamous cell carcinoma had an elevated serum HGF compared to patients with cervical intraepithelial neoplasia or healthy controls (Zhang et al. 2014). Furthermore, HGF serum concentrations were related to the pathological grade, International Federation of Gynecology and Obstetrics (FIGO) stage, pelvic lymph node metastasis as well as tumor size. Interestingly, the serum levels of HGF were also reported to be higher in patients with metastasis compared to those without metastasis, suggesting that HGF expression may be correlated with cancer cell metastasis and infiltration. However, the latter did not reach statistical significance.

Refaat and al assessed retrospectively the association between pre-treatment c-MET expression and treatment outcome in 28 patients treated for an advanced CC with concurrent chemo-radiation therapy (Refaat et al. 2017). They showed an association between c-MET overexpression and poor OS, PFS, distant metastasis control and loco-regional control.

A meta-analysis (Peng et al. 2016) showed that c-Met expression was higher in cervical cancer (60.99%) than in non-neoplastic cervical tissue (19.74%). Results showed that increasing malignancy resulted in elevated c-Met expression. The relationship between c-Met expression and clinic-pathologic features was also evaluated. c-Met expression was correlated with disease-free survival, lymph node involvement, and lymphovascular space invasion.

However, the molecular mechanisms underlying the upregulation of this proto-oncogene are unknown. Recently, Chen and al found a relationship between the oncogenic E6 proteins expressed by high-risk HPV isotypes and epigenetic activation of super-enhancers in the genome that drive expression of key oncogenes like *EGFR* and *c-MET* (Chen et al. 2018).

A study showed that downregulation of MET receptor expression via RNA interference in different cervical carcinoma cell lines dramatically decreased tumor growth and forced tumor differentiation *in vivo*, decreased the expression of a key regulator of the epithelial-to-mesenchymal transition, SLUG, and increased the expression of E-cadherin, a hallmark of the epithelial phenotype and impairs expression and signaling of CXCR4 receptor, responsible for invasive phenotype (Miekus et al. 2015). MET receptor silencing also led to a dramatic decrease in cell size and a decrease in proliferation rate under normal and stress conditions.

These findings highlight the role of the MET receptor in cervical carcinoma cells and indicate the MET receptor as a potential therapeutic target for advanced cervical carcinoma.

3.2 CABOZANTINIB INFORMATION

Cabozantinib is an oral small molecule tyrosine kinase inhibitor that targets VEGF receptor as MET and AXL. It is currently used to treat hepatocellular carcinoma (Elisei et al. 2013) and metastatic renal cell carcinoma (Choueiri et al. 2015). Its angiogenic properties associated with MET and AXL inhibition represent a potential active treatment in cervical carcinoma.

Cabozantinib exhibits potent inhibitory activity against several receptor tyrosine kinases that are known to influence tumor growth, metastasis, and angiogenesis. The primary targets of cabozantinib are VEGFR2/KDR, MET and RET with cell-based IC₅₀ (concentration associated with 50% inhibition) values of 2, 8 and 85 nM, respectively. In addition, cabozantinib inhibited phosphorylation of KIT, FLT3, and AXL with IC₅₀ values of 5, 11, and 42 nM, respectively.

The clinical activity of cabozantinib has been evaluated in pivotal trials in renal cell carcinoma (RCC), medullar thyroid carcinoma, hepatocellular carcinoma and castration resistant prostate cancer. In Phase I and Phase II studies, cabozantinib has also demonstrated findings consistent with broad clinical antitumor activity in multiple tumor types.

A randomized, open-label, active-controlled phase III study XL184-308 (METEOR trial) was conducted in 658 subjects (330 cabozantinib, 328 everolimus) with advanced RCC who had received prior treatment with at least one VEGFR-TKI (Choueiri et al. 2016). Cabozantinib demonstrated statistically significant improvements in the primary endpoint (PFS) and both secondary endpoints (ORR, OS) compared with the standard-of-care control treatment (everolimus). In the PFS analysis performed in the first 375 randomized patients, the multivariable hazard ratio (HR) was 0.58 (95% CI: 0.45, 0.74; p-value < 0.0001), and the Kaplan-

Meier estimates for median duration of PFS were 7.4 months in the cabozantinib arm versus 3.8 months in the everolimus arm. They further observed significantly consistent prolongation of OS with a minimum follow-up of 13 months in the cabozantinib versus everolimus arm with an adjusted HR, of 0.66 (95% CI: 0.53, 0.83; $p = 0.0003$). Kaplan-Meier estimates for median duration of OS were 21.4 months in the cabozantinib arm and 16.5 months in the everolimus arm. The most common grade 3 or 4 adverse events in the cabozantinib group were hypertension (15%), diarrhea (13%), fatigue (11%), palmar-plantar erythrodysesthesia syndrome (8%), anaemia (6%), hyperglycaemia (1%) and hypomagnesaemia (5%) which are common side effects of antiangiogenic treatments.

3.3 STUDY RATIONALE AND PURPOSE

It therefore appears challenging to assess the interest of cabozantinib among patients with advanced/metastatic cervical cancer where few therapeutics options are available after failure to platinum-based regimen.

Given the previous findings with other antiangiogenic drugs with regards to tolerance profile, a specific attention had to be paid on toxicities under cabozantinib treatment.

In this context, we propose to implement a phase II trial aiming to simultaneously assess efficacy and safety of cabozantinib monotherapy in advanced/metastatic cervical carcinoma after failure to platinum-based regimen treatment, as proposed by the Bryant and Day design (Bryant & Day 1995).

3.4 BIOLOGICAL ANCILLARY STUDY

3.4.1 Biomarkers

Development of cervical carcinoma is a multistep process initiated by persistent infection with high-risk HPV. The most important prognostic variables of cervical carcinoma are represented by clinical or histological data (FIGO stage, lymph-vascular space involvement). Recently, there has been increasing interest in the identification of biomarkers able to predict both response to treatment and survival.

Among the different biomarkers, we will be interested in immunological and angiogenesis biomarkers. Cabozantinib is a multi-target tyrosine kinase inhibitor, including VEGFR, MET, RET, KIT, AXL, FLT3 and TIE-2. Aberrant MET activation is known to promote tumor growth, anti-apoptosis, angiogenesis, invasion, and metastasis (Gherardi et al. 2012). In CABOSUN trial and in METEOR trial, MET surexpression was defined as high versus low, based on a cutoff of 50% or higher of tumour tissue stained with an intensity of 2+ or 3+ by immunohistochemistry (Choueiri et al. 2016). Results suggest that MET expression level is a prognostic and a predictive marker of cabozantinib in renal cell carcinoma. Moreover, VEGF, already known for its major role in neoangiogenesis, was recently identified as a key factor in tumor-induced immunosuppression (Lapeyre-Prost et al. 2017; Kwilas et al. 2014). In particular, VEGF-A fosters the proliferation of immunosuppressive cells, limits T-cell recruitment into tumors, and promotes T-cell exhaustion (LAPEYRE et KWILAS). Recent data suggest the immune modulatory role of cabozantinib (Patnaik et al 2017).

3.4.2 Monitoring of tumoral response

Circulating cell-free tumoral DNA (ctDNA) has been widely evaluated as a liquid biopsy for detecting cancer, monitoring disease, characterizing drug targets, and uncovering resistance in various tumors. In HPV-related cancer, HPV viral genomes are usually integrated into the tumor cell genome or episomal DNA. Droplet-digital PCR is able to detect and quantify tumor-derived HPV DNA sequences in patient blood (HPV ctDNA) with high sensitivity and specificity (Campitelli et al. 2012; Jeannot et al. 2016).

3.4.3 Sarcopenia induced by cabozantinib

Cabozantinib is an inhibitor of tyrosine kinases including VEGFR, cMET, and AXL4, currently approved in metastatic renal cell carcinoma. In METEOR pivotal trial, cabozantinib is associated with 68% of grade 3-4 adverse events (Choueiri et al. 2015 ; Choueiri et al. 2017). More than one-third of cancer patients experiences weight loss of more than 5% during the disease (Tisdale et al. 2002).

More than half of cancer patients die with cachexia (Fearo et al, 2011). Weight loss leads to impaired physical performance resulting in a continuous deterioration of the patient's overall state and well-being (Arends et al, 2017). Indeed, weight loss influences the diagnosis of malnutrition, that is provided on the basis of a full medical examination with biologicals and clinical parameters. Associated to impaired quality of life, decreased overall survival, and increased toxicities from oncologic treatment, malnutrition is a serious concern in oncology (Morel et al. 2018 ; Chemama et al. 2016). The global incidence of malnutrition in cancer ranges from 30% to 85%, being most prevalent in patients with gastric, pancreatic, lung head and neck, lung, and colon cancer (Arends et al. 2017 ; Laviano et al. 1996 ; Bozzetti et al. 2001) . In metastatic cervical carcinoma, malnutrition is a real concern. Sarcopenia is a syndrome characterized by low skeletal muscle mass with impaired muscle function, with multifactorial etiology. Described with antiangiogenics, sarcopenia is related to higher toxicity and/or poor responses to antineoplastic drugs, and decreased survival in cancer patients (Davis et al. 2019 ; Fukushima et al. 2016). For examples in metastatic renal cell carcinoma, half of patients had sarcopenia before starting first line sorafenib, and antiangiogenics used to exacerbate muscle mass wasting (Antoun et al. 2010). Indeed, usuals antiangiogenics gastro-intestinal side effects are characterized by diminished nutrient intake and progressive tissue depletion. By the way, antiangiogenics used to induce weight loss with G3-4 around 1-2% in trials, due to direct effect on muscle tissue also (Zhang et al. 2014). Indeed, there is evidence that these drugs promote reduced skeletal muscle maturation and growth by molecular pathways (Huot et al. 2019). Muscle mass wasting is reported under antiangiogenics (Huillard et al. 2013 ; Phillips et al. 2019). Based on our experience in metastatic renal cell carcinoma, 70% of patients at baseline are sarcopenic; and half of patients without sarcopenia at baseline became sarcopenic under cabozantinib. As cabozantinib is now broadly prescribed, it is necessary to anticipate and investigate this underreported side effect.

CABOCOL trial could investigate prospectively the impact of cabozantinib on weight loss and sarcopenia in an homogenous population with high risk of malnutrition.

The aim of this ancillary study was to explore sarcopenia under cabozantinib to evaluate the impact on patient's outcomes in advanced and metastatic cervical carcinoma.

4 STUDY OBJECTIVES

4.1 PRIMARY OBJECTIVE

The primary objective is to assess efficacy and safety of cabozantinib in monotherapy in advanced/metastatic cervical cancer (CC) after failure of platinum-based regimen treatment.

4.2 SECONDARY OBJECTIVES

The secondary objectives are:

- ◆ To assess:
 - The safety profile of cabozantinib
 - The objective response (RECIST v1.1 criteria)
 - The progression-free and overall survival
 - The health-related quality-of-life (QoL) of patients
 - The nutritional outcomes
- ◆ To constitute biological collections (blood, tumoral tissue) for further biological explorations such as:
 - Evaluation of the impact of immune and angiogenesis biomarkers and intra-cellular activated pathways (blood, ctDNA and tissue) in terms of oncological outcomes and response to treatment,
 - Exploration of the modulatory role of cabozantinib on immune cells,
 - Evaluation of the predictive value of HPV-ctDNA during cabozantinib treatment

5 ENDPOINTS

5.1 PRIMARY ENDPOINT

The main objective will be based on **joint primary endpoints of efficacy and safety** as proposed by the Bryant-and-Day design:

- **Efficacy:** assessed by the proportion of patients with disease control rate 3 months after cabozantinib treatment initiation.
- **Safety** assessed by the proportion of patients with clinical gastro-intestinal (GI) perforation/fistula, GI-vaginal fistula and genito-urinary (GU) fistula events grade ≥ 2 (NCI CTCAE V.5) occurring up to 1 month after the end of treatment

5.2 SECONDARY ENDPOINTS

The secondary endpoints are:

- Progression-free survival, defined as the time between initiation of cabozantinib treatment and progression (RECIST 1.1 criteria) or death of any cause whichever occurs first.
- Objective response rate defined as the percentage of patients who have achieved complete response or partial response with RECIST 1.1 criteria.
- Overall survival, defined as the time between initiation of cabozantinib treatment and death of whatever cause.
- Toxicities evaluated according to NCI CTCAE v5.0 criteria, in terms of kind, grade, time of onset, reversibility
- Significant toxicities responsible for interruption or dose reduction of cabozantinib treatment
- Proportion of patients requiring cabozantinib dose reduction or treatment stopping for toxicity
- Scores of Quality-of-life according to French versions of the self-administered standardized validated questionnaire: EORTC QLQ-C30 and its additional cervix cancer module (QLQ-CX24)
- To investigate nutritional outcomes, including: weight, sarcopenia, biological markers, muscle mass, and muscle strength under cabozantinib in advanced cervical carcinoma

6 STUDY DESIGN

6.1 METHODOLOGY

We propose a phase II single-arm two-stage multicentre study aiming to simultaneously assess efficacy and safety of cabozantinib.

6.2 STUDY DURATION

- Start of study: 4th Trimester 2019
- Total inclusion period : 30 months
- Total study period : 42 months (30 months and 1 year for follow-up)

6.3 SUBJECTS' SELECTION

6.3.1 Inclusion criteria

- Female 18 years of age or older
- Histologically confirmed recurrent unresectable, persistent or metastatic cervix carcinoma with squamous cell, adenocarcinoma or adenosquamous histology.
- Patient may have received at least one prior chemotherapy regimen of platinum-based chemotherapy for recurrence or metastatic disease. Cisplatin given in combination with radiation for a localized disease does not count as a prior chemotherapy. Prior treatment for advanced/metastatic disease with bevacizumab is allowed. Prior treatments with immune checkpoint inhibitors are allowed.
- ECOG performance status 0-2

- Measurable disease per RECIST 1.1
- The subject must have recovered to baseline or CTCAE v.5.0 (Common Terminology Criteria for Adverse Events, version 5.0) $\leq$ Grade 1 from clinical toxicities related to any prior treatments, i.e chemotherapy or pelvis radiation unless AE(s) are clinically non-significant (for example alopecia)
- Adequate organ and marrow function, defined as follows, based upon laboratory tests performed within 7 days before inclusion:
 - o Absolute neutrophil count (ANC) $\geq 1000/\text{mm}^3$ ($\geq 1.0 \text{ GI/L}$)
 - o Platelets $\geq 100,000/\text{mm}^3$ ($\geq 100 \text{ GI/L}$)
 - o Hemoglobin $\geq 10 \text{ g/dL}$ ($\geq 100 \text{ g/L}$) (red blood cell transfusion is allowed)
 - o Total bilirubin ≤ 1.5 fold the upper limit of normal (for subjects with Gilbert's disease, $\leq 3 \text{ mg/dL}$ or $\leq 51.3 \mu\text{mol/L}$)
 - o Serum albumin $\geq 3.0 \text{ g/dL}$ ($\geq 30 \text{ g/L}$)
 - o Calculated creatinine clearance $\geq 30 \text{ mL/min}$ by CKD-EPI method
 - o Alanine aminotransferase (ALT) and aspartate aminotransferase (AST) $< 3.0 \times$ the upper limit of normal
 - o Urine protein/creatinine ratio (UPCR) $\leq 1\text{g/g}$ ($\leq 113.17 \text{ mg/mmol creatinine}$) or 24-hour urine protein $< 1 \text{ g}$
- Left-ventricular ejection fraction $\geq 50\%$
- Subjects affiliated to an appropriate social security system
- Female subjects of childbearing potential must not be pregnant at screening and during treatment by Cabozantinib. Effective methods of contraception should be used throughout the course of treatment and for at least 4 months after the end of treatment. Sexually active fertile subjects and their partners must agree to use medically accepted barrier methods of contraception (e.g., male or female condom) during the study and 4 months after the last dose of study treatment, even if oral contraceptives are also used.
- Before patient registration, written informed consent must be given according to ICH/GCP, and national/local regulations.

6.3.2 Non inclusion criteria

- Clinically significant gastrointestinal abnormalities that may increase the risk for gastrointestinal bleeding and/or fistula including, but not limited to: Active peptic ulcer disease, inflammatory bowel disease (e.g. ulcerative colitis, Crohn's disease), or other gastrointestinal conditions with increased risk of perforation, history of abdomino and/or pelvic fistula, gastrointestinal perforation, or intra-abdominal abscess, gastro-intestinal obstruction
- Patients with lesions on baseline pelvic MRI which may major the risk of abdominal and/or pelvic fistula/perforation
- Clinically significant gastrointestinal abnormalities that may affect absorption of investigational product including, but not limited to: malabsorption syndrome, major resection of the stomach or small bowel.
- Previously identified allergy or hypersensitivity to components of the study treatment formulations (Note: patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take cabozantinib and are also excluded).
- History of any one or more of the following cardiovascular conditions within the past 6 months: Cardiac angioplasty or stenting, myocardial infarction, unstable angina, coronary artery bypass surgery, symptomatic peripheral vascular disease, class III or IV congestive heart failure, as defined by the New York Heart Association (NYHA), serious cardiac arrhythmias.
- Corrected QT interval (QTc) calculated by the Fridericia formula $> 500 \text{ msec}$ within 28 days before inclusion (see Annex for Fridericia formula). Note: if initial QTcF is found to be $> 500 \text{ msec}$, two additional ECGs separated by at least 3 minutes should be performed. If the average of these three consecutive results for QTcF is $\leq 500 \text{ msec}$, the subject meets eligibility in this regard.
- Uncontrolled hypertension defined as systolic blood pressure (SBP) of $> 150 \text{ mmHg}$ or diastolic blood pressure (DBP) of $> 100 \text{ mmHg}$ despite an optimal treatment.

- History of cerebrovascular accident including transient ischemic attack (TIA), symptomatic pulmonary embolism or untreated deep venous thrombosis (DVT) within the past 6 months. *Note: Subjects with recent DVT or asymptomatic pulmonary embolism who have been treated with therapeutic anti-coagulating agents for at least 4 weeks are eligible.*
- Major surgery or trauma within 28 days prior to first dose of investigational product and/or presence of any non-healing wound, fracture, or ulcer.
- Evidence of active bleeding or pathologic conditions that carry high risk of bleeding such as coagulopathy or tumor involving major vessels.
- At least 6 weeks must have elapsed between the last dose of pelvis palliative radiation and the first dose of cabozantinib or 2 weeks for other localization of palliative radiation
- Presence of brain metastases or epidural disease unless adequately treated with radiotherapy and/or surgery (including radiosurgery) and stable for at least 3 months before inclusion. Eligible subjects must be neurologically asymptomatic and without corticosteroid treatment at the time of inclusion.
- Concomitant use of known strong CYP3A4 inhibitors or inducers.
- Patients with second primary cancer, except adequately treated non-melanoma skin cancer, or other solid tumors curatively treated with no evidence of disease for ≥ 3 years
- Any psychological, familial, sociological or geographical condition potentially hampering compliance with the study protocol and follow-up schedule; those conditions should be discussed with the patient before registration in the trial
- Concurrent participation in any therapeutic clinical trial
- Patient deprived of liberty or placed under the authority of a tutor

6.4 STUDY PLAN

6.4.1 Consent sign

The study will be proposed by oncologists to patients meeting the eligibility criteria. Patients will be given study information (explanation on the study, reading and giving of the information notice). Patients will have a reflection period of the duration of their choice.

After collection of the signed informed consent, the selection criteria will be verified before inclusion in the trial. The specific exams/procedure requested in the study before inclusion (baseline visit) will be realized after signed consent and before inclusion.

6.4.2 Inclusion procedure

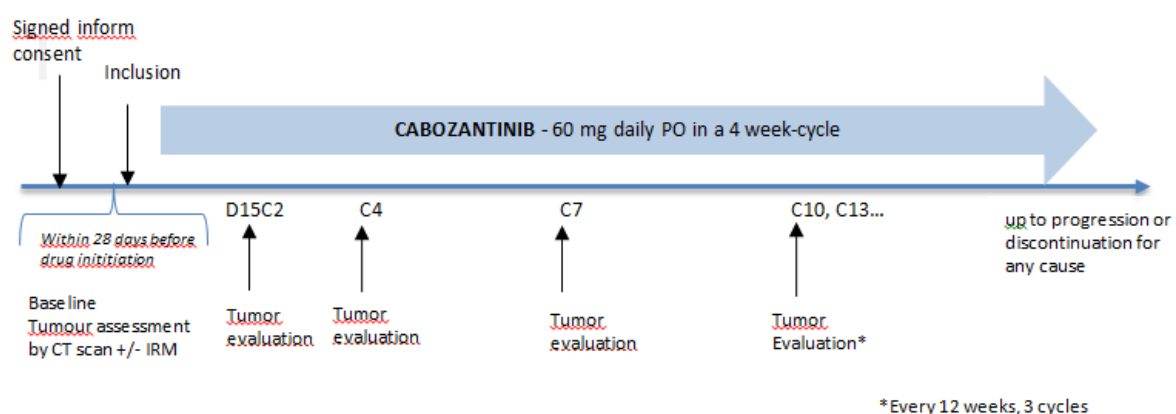
After collection of the signed informed consent and checking of all eligibility criteria, inclusion will be performed before initiation of treatment.

The inclusion will be recorded on the software dedicated to the study via an internet portal.

An inclusion number will be assigned to the patient and will be used during the whole study.

6.4.3 Drug administration

Eligible patients who have completed screening and have signed the informed consent to participate to this two-step phase II trial will receive Cabozantinib treatment.



Cabozantinib will be administered at the daily dose of 60 mg given orally in a 4-week cycle. It will be continued without interruption until disease progression or discontinuation for any cause.

Cabozantinib must be taken on an empty stomach. Patients must fast for 2 hours before and 1 hour following each dose of Cabozantinib. Cabozantinib tablets should be swallowed whole with at least 240 mL of water.

The tablets should not be crushed. Subjects should be instructed to take their cabozantinib dose at approximately the same time every day. If a subject misses a dose, the dose may be taken later only if it is within 12 hours of when the missed dose should have been taken. The missed dose should not be made up if it is within 12 hours of the next scheduled dose.

In case of vomiting, patients will be instructed to not take a new dose.

Grapefruit, grapefruit juice, Seville oranges and their products should be avoided by subjects taking cabozantinib.

Patients will be instructed to bring all unused tablets and their medication diary to each study visit for assessment of compliance.

6.4.4 Premedication

No premedication is required in this study.

6.5 SCHEDULE OF ASSESSMENT

6.5.1 Inclusion Assessments

Before treatment start, the following evaluations will be performed within 28 days before cabozantinib initiation for eligible patients with signed informed consent:

- ✓ Clinical assessment : complete physical examination including gynecological examination, weight, height, ECOG, vital signs
- ✓ Laboratory Assessments (within 14 days before cabozantinib initiation):
 - Hematology (CBC, platelets)
 - Serum biochemistry (Albumin, total alkaline phosphatase (ALP), alanine amino transferase (ALT), aspartate amino transferase (AST), corrected calcium, creatinine and clearance by CKD-EPI method, glutamyltransferase (GGT), glucose, lactate dehydrogenase (LDH), magnesium, phosphorus, potassium, sodium, total bilirubin (conjugated and unconjugated if clinically indicated), total protein)
 - Coagulation (PT/INR and PTT)
 - Thyroid function tests (TSH, free T3, free T4)
 - Tumor marker : SCC antigen
 - Urine analysis: urine protein-to-creatinine ratio (UPCR) ≤ 1 g/g (≤ 113.2 mg/mmol) creatinine or 24-hour urine protein < 1 g. When a UPCR exceeds 1g/g, a repeat UPCR **or** a 24-hour urine protein and creatinine should be performed to confirm the result.
 - Urinal pregnancy test (women of childbearing potential)
- ✓ Electrocardiogram (ECG) to determine the corrected QT interval
- ✓ Echography or MUGA
- ✓ Quality-of-life assessment using the self-questionnaires :
 - EORTC QLQ-C30 and QLQ-CX24
- ✓ Radiological assessment:
 - CT scan of the thorax, abdominal and pelvis using RECIST version 1.1.
 - pelvis MRI (precaution : patients with lesions on baseline pelvic MRI which may major the risk of abdominal and/or pelvic fistula/perforation should not be included)
 - PET-CT is optional
- ✓ Tumoral biopsy from primitive and/or from metastatic sites if feasible for biological collection (optional, if feasible and if the patient agrees on the consent form)
- ✓ Blood samples for biological collection

6.5.2 Assessments during treatment

Day 1 of cycle 1

- ✓ Clinical assessment : complete physical examination including gynecological examination, weight, ECOG, vital signs
- ✓ Laboratory Assessments (only if realized more 14 days before D1) :
 - Hematology (CBC, platelets)
 - Serum biochemistry (Albumin, total alkaline phosphatase (ALP), alanine amino transferase (ALT), aspartate amino transferase (AST), corrected calcium, creatinine and clearance by CKD-EPI method, glutamyltransferase (GGT), glucose, lactate dehydrogenase (LDH), magnesium, phosphorus, potassium, sodium, total bilirubin (conjugated and unconjugated if clinically indicated), total protein)
 - Coagulation (PT/INR and PTT)
 - Thyroid function tests (TSH, free T3, free T4)
 - Tumor marker : SCC antigen
 - Urine dipstick (with protein, blood and leucocytes)
If results > 2+, an urinary analysis must be performed with a Urine protein / creatinine ratio (UPCR). When a UPCR exceeds 1 g/g, a repeat UPCR or a 24-hour urine protein and creatinine should be performed to confirm the result.
- ✓ Diary card to give to the patient

Subjects should be instructed before the beginning of the treatment of the risk of diarrhea and the initial management. A preventive prescription can be given to the patient at the beginning of the treatment.

D15 of cycle 1 (+/- 1 day):

- ✓ Clinical assessment : complete physical examination including gynecological examination if clinically indicated, weight, ECOG, vital signs, adverse events, concomitant treatments)
- ✓ Laboratory Assessments (may be done within 3 days before D15)
 - Hematology (CBC, platelets)
 - Serum biochemistry (Albumin, total alkaline phosphatase (ALP), alanine amino transferase (ALT), aspartate amino transferase (AST), corrected calcium, creatinine and clearance by CKD-EPI method, glutamyltransferase (GGT), glucose, lactate dehydrogenase (LDH), magnesium, phosphorus, potassium, sodium, total bilirubin (conjugated and unconjugated if clinically indicated), total protein)
 - Coagulation (PT/INR and PTT) if clinically indicated
 - Urine dipstick (with protein, blood and leucocytes)
If results > 2+, an urinary analysis must be performed with a Urine protein / creatinine ratio (UPCR). When a UPCR exceeds 1 g/g, a repeat UPCR or a 24-hour urine protein and creatinine should be performed to confirm the result.

Day 1 of each cycle from cycle 2 (+/- 1 day):

- ✓ Clinical assessment : complete physical examination including gynecological examination, weight, ECOG, vital signs, adverse events, concomitant treatments
- ✓ Laboratory Assessments (may be done within 3 days before D1):
 - Hematology (CBC, platelets)
 - Serum biochemistry (Albumin, total alkaline phosphatase (ALP), alanine amino transferase (ALT), aspartate amino transferase (AST), corrected calcium, creatinine and clearance by CKD-EPI method, glutamyltransferase (GGT), glucose, lactate dehydrogenase (LDH), magnesium, phosphorus, potassium, sodium, total bilirubin (conjugated and unconjugated if clinically indicated), total protein)
 - Coagulation (PT/INR and PTT) if clinically indicated
 - Thyroid function tests (TSH, free T3, free T4)
 - Tumor marker : SCC antigen
 - Urine dipstick (with protein, blood and leucocytes)
If results > 2+, an urinary analysis must be performed with a Urine protein / creatinine ratio (UPCR). When a UPCR exceeds 1 g/g, a repeat UPCR or a 24-hour urine protein and creatinine should be performed to confirm the result.
- ✓ Electrocardiogram (ECG) to determine the corrected QT interval (only at D1C2,C4,C7...)
- ✓ Diary card to give to the patient
- ✓ Blood samples for biological collection

D15 of cycle 2 (+/- 1 day):

- ✓ Clinical assessment : complete physical examination including gynecological examination if clinically indicated, weight, ECOG, vital signs, adverse events, concomitant treatments
- ✓ Laboratory Assessments (may be done within 3 days before D15):
 - Hematology (CBC, platelets)
 - Serum biochemistry (Albumin, total alkaline phosphatase (ALP), alanine amino transferase (ALT), aspartate amino transferase (AST), corrected calcium, creatinine and clearance by CKD-EPI method, glutamyltransferase (GGT), glucose, lactate dehydrogenase (LDH), magnesium, phosphorus, potassium, sodium, total bilirubin (conjugated and unconjugated if clinically indicated), total protein)
 - Coagulation (PT/INR and PTT) if clinically indicated
 - Tumor marker : SCC antigen
 - Urine dipstick (with protein, blood and leucocytes)
If results > 2+, an urinary analysis must be performed with a Urine protein / creatinine ratio (UPCR). When a UPCR exceeds 1 g/g, a repeat UPCR or a 24-hour urine protein and creatinine should be performed to confirm the result.
- ✓ Radiological assessment (will use the same exams as baseline) may be done within 3 days before D15:
 - CT scan of the thorax, abdominal and pelvis using RECIST version 1.1.
 - Optional pelvis MRI can be used in addition of CT scan if a local recurrence could not be assessed by CT
- ✓ Quality-of-life assessment using the self-questionnaires :
 - EORTC QLQ-C30 and QLQ-CX24
- ✓ Blood samples for biological collection

Additional exams at D1 of cycle 4 and every 3 cycles (D1C7, D1C10...) (within 7 days):

- ✓ Radiological assessment (will use the same exams as baseline)
 - CT scan of the thorax, abdominal and pelvis using RECIST version 1.1.
 - Optional pelvis MRI can be used in addition of CT scan if a local recurrence could not be assessed by CT
- ✓ Quality-of-life assessment using the self-questionnaires :
 - EORTC QLQ-C30 and QLQ-CX24
- ✓ Electrocardiogram (ECG) to determine the corrected QT interval

6.5.3 End of treatment assessments

Patients will be followed-up 30 days after the last dose of study treatment (+/- 7 days):

- ✓ Clinical assessment : complete physical examination including gynecological examination, weight, ECOG, vital signs, adverse events, concomitant treatments
- ✓ Laboratory Assessments:
 - Hematology (CBC, platelets)
 - Serum biochemistry (Albumin, total alkaline phosphatase (ALP), alanine amino transferase (ALT), aspartate amino transferase (AST), corrected calcium, creatinine and clearance by CKD-EPI method, glutamyltransferase (GGT), glucose, lactate dehydrogenase (LDH), magnesium, phosphorus, potassium, sodium, total bilirubin (conjugated and unconjugated if clinically indicated), total protein)
 - Coagulation (PT/INR and PTT) if clinically indicated
 - Thyroid function tests (TSH, free T3, free T4)
 - Tumor marker : SCC antigen
 - Urine dipstick (with protein, blood and leucocytes)
If results > 2+, an urinary analysis must be performed with a Urine protein / creatinine ratio (UPCR). When a UPCR exceeds 1 g/g, a repeat UPCR or a 24-hour urine protein and creatinine should be performed to confirm the result.

- ✓ Radiological assessment (will use the same exams as baseline)
 - CT scan of the thorax, abdominal and pelvis using RECIST version 1.1.
 - Optional pelvis MRI can be used in addition of CT scan if a local recurrence could not be assessed by CT
- ✓ Quality-of-life assessment using the self-questionnaires :
 - EORTC QLQ-C30 and QLQ-CX24
- ✓ Blood samples for biological collection (if disease progression)

6.5.4 Follow-up assessments

Patients who discontinue the treatment for another reason than disease progression will be followed-up every 3 months with:

- ✓ Clinical assessment : complete physical examination including gynecological examination, weight, ECOG, vital signs, adverse events, concomitant treatments
- ✓ Laboratory Assessments only if biological toxicity not recovered at the end of treatment:
 - Hematology (CBC, platelets)
 - Serum biochemistry (Albumin, total alkaline phosphatase (ALP), alanine amino transferase (ALT), aspartate amino transferase (AST), corrected calcium, creatinine and clearance by CKD-EPI method, glutamyltransferase (GGT), glucose, lactate dehydrogenase (LDH), magnesium, phosphorus, potassium, sodium, total bilirubin (conjugated and unconjugated if clinically indicated), total protein)
 - Coagulation (PT/INR and PTT) if clinically indicated
 - Thyroid function tests (TSH, free T3, free T4)
 - Tumor marker : SCC antigen
- ✓ Urine dipstick (with protein, blood and leucocytes)
 - If results > 2+, an urinary analysis must be performed with a Urine protein / creatinine ratio (UPCR). When a UPCR exceeds 1 g/g, a repeat UPCR or a 24-hour urine protein and creatinine should be performed to confirm the result.
- ✓ Radiological assessment (will use the same exams as baseline)
 - CT scan of the thorax, abdominal and pelvis using RECIST version 1.1.
 - Optional pelvis MRI can be used in addition of CT scan if a local recurrence could not be assessed by CT
- ✓ Quality-of-life assessment using the self-questionnaires :
 - EORTC QLQ-C30 and QLQ-CX24
- ✓ Blood samples for biological collection (if disease progression)

For patients withdrawn from treatment due to progression, a follow-up for survival will be collected in this study. No study visit is required. The following treatment is at the discretion of physician.

6.6 CRITERIA FOR PREMATURE WITHDRAWAL

The treatment will be interrupted at any time under the following circumstances:

- Disease progression
- Need to initiate another anti-tumor treatment (e.g., systemic anticancer treatment palliative radiation and surgery)
- Unacceptable toxicity, not compatible with study treatment
- Necessity for withholding study drug for greater than 4 weeks for study-treatment related AEs.
- Significant noncompliance with the protocol schedule in the opinion of the investigator or the Sponsor.
- Patient's decision (the data already collected during the search can be kept and exploited unless the patient opposes it)
- Intercurrent illness or other reason that requires stopping treatment of the study
- Patient lost to view
- Investigator's decision

6.7 CONCOMITANT MEDICATION

Because there is a potential for interaction of Cabozantinib with other concomitantly administered drugs through the cytochrome P450 system, the case report form must capture the concurrent use of all other drugs, over-the-counter medications, or alternative therapies. The Investigator should be alerted if the patient is taking any agent known to affect or with the potential to affect selected CYP450 isoenzymes.

The following treatments/procedures are permitted:

- **Antiemetics and antidiarrheal** medications are allowed prophylactically in accordance to standard clinical practice if clinically indicated;
- **Transfusions, hormone replacement, and corticosteroids** should be used as indicated by standard clinical;
- Individualized **anticoagulation therapy** with heparin is allowed if it can be provided safely and effectively under the following circumstances:
 - Low dose heparins for prophylactic use are allowed if clinically indicated and the benefit outweighs the risk per the investigator's discretion;
 - Therapeutic doses of low molecular weight heparin (LMWH) are allowed if clinically indicated (eg, for the treatment of deep venous thrombosis), and the benefit outweighs the risk per the investigator's discretion.

For management of thromboembolic complications while on study, accepted clinical guidelines regarding appropriate management while receiving anticoagulation therapy with heparins must be followed. This includes, but is not limited to, subject education regarding potential adverse drug reactions, monitoring laboratory parameters, dose adjustments (eg, due to kidney dysfunction).

- **Antacids, H2-blockers, PPIs:** Co-administration of gastric pH modifying drugs such as PPI, H2-blockers or antacids is not contraindicated in subjects administered cabozantinib.

The following treatments are not permitted:

- **Anticancer therapy.** If a subject requires additional systemic anticancer treatment, study treatment must be discontinued. Local intervention is discouraged unless medically unavoidable. Subjects receiving local intervention (e.g., palliative radiation and surgery) are not allowed to continue to receive study treatment.
- **Therapeutic doses of oral anticoagulants** (eg, warfarin or other coumarin-related agents, direct thrombin or direct FXa inhibitors). They should be replaced with therapeutic doses of LMWH
- Bisphosphonates/Denosumab

The following treatments should be avoided if possible, while the subject is on study:

- Palliative external radiation should not be performed while on study.
- Erythropoietic stimulating agents should not be used based on a report of increased risk of tumor recurrence and/or progression associated with erythropoietin;
- Chronic co-administration of cabozantinib with strong inducers of the CYP3A4 family (eg, dexamethasone, phenytoin, carbamazepine, rifampin, rifabutin, rifapentine, phenobarbital, and St. John's Wort) may significantly decrease cabozantinib concentrations and should be avoided. Selection of alternate concomitant medications with no or minimal CYP3A4 enzyme induction potential is recommended;
- Co-administration of cabozantinib with strong inhibitors of the CYP3A4 family (eg, ketoconazole, itraconazole, clarithromycin, indinavir, nefazodone, nelfinavir, and ritonavir) may increase cabozantinib concentrations and should be avoided. Grapefruit and Seville oranges may also increase plasma concentrations of cabozantinib and should be avoided.
- Drugs Associated with QTc Prolongation: Treatment with cabozantinib has been associated with a mild prolongation of the QTc interval. Caution should be used when treating subjects on cabozantinib with other drugs associated with QTc prolongation. Additional QTc monitoring is suggested for subjects who are treated concomitantly with QTc prolonging drugs.
- In case of dental surgery or invasive dental treatment, cabozantinib should be suspended for at least 28 days before surgery. Therefore, the patient should be withdrawn of study.

7 PHARMACEUTICAL FORM OF STUDY MEDICATION

7.1 CABOZANTINIB

Cabozantinib will be supplied by Ipsen laboratory.

7.1.1 Presentation of the product

The commercial name of the study treatment is Cabometyx.

7.1.2 Packaging and Labelling

Cabozantinib will be provided as tablets of 60 mg, 40 mg and 20 mg.

7.1.3 Storage conditions

The material is not considered hazardous and the allowable storage condition is room temperature.

7.1.4 Method of administration

Cabozantinib will be administered orally with initial dose of 60 mg/day.

Cabozantinib must be taken on an empty stomach. Patients must fast for 2 hours before and 1 hour following each dose of Cabozantinib. Cabozantinib tablets should be swallowed whole with at least 240 mL of water.

The tablets should not be crushed. Subjects should be instructed to take their cabozantinib dose at approximately the same time every day. If a subject misses a dose, the dose may be taken later only if it is within 12 hours of when the missed dose should have been taken. The missed dose should not be made up if it is within 12 hours of the next scheduled dose. In case of vomiting, patients will be instructed to not take a new dose. Grapefruit, grapefruit juice, Seville oranges and their products should be avoided by subjects taking cabozantinib.

Patients will be instructed to bring all unused tablets and their medication diary to each study visit for assessment of compliance.

7.1.5 Label

The product will be labeled according to current legal requirements.

8 DOSE ADJUSTMENTS OF STUDY PRODUCT

8.1 GENERAL RULES

- Once doses are reduced, **no dose increase** will be possible.
- Any patients requiring more than 2 dose reductions of Cabozantinib will be withdrawn from the study.
- The maximum interruption **allowed is 28 successive days** for the study treatment. Beyond this time, the patient will be withdrawn from the study treatment.

8.2 DOSE LEVELS

Dose level	Starting dose Level 0	First level reduction Level-1	Second level reduction Level-2
CABOZANTINIB	60 mg/day	40 mg/day	20 mg/day

8.3 ADJUSTMENT

The following guideline explains dose adjustments for several of these toxic effects. If a patient experiences several adverse events and there are conflicting recommendations, the recommended dose adjustment that reduces the dose to the lowest level will be used.

Guidelines for AE management and treatment modifications are outlined below. *Unless otherwise specified in the protocol, these parameters are only guidelines and are not intended to supersede the clinical judgment of the treating physician.*

8.3.1 Fistula/perforation

If a patient develops a symptomatic gastro-intestinal, vaginal or urinary fistula / perforation, cabozantinib will be discontinued and MRI scan, and biopsy if clinically applicable, will be realized, to determine whether the fistula is associated with PD. Permanently discontinue cabozantinib in patients with $\geq$ grade 2 fistula. If the physician considers it to be in the best interests of the patient to continue cabozantinib, the physician must submit a request to the sponsor, which will review the case.

Each case of fistula / perforation will have to be notified as an AE of Special Interest (AESI), whatever the grade, on SAE form.

8.3.2 Hematologic adverse event

Table 1. General approach for the management of cabozantinib-related hematologic adverse event	
CTCAE V.5.0	Guidelines/Intervention
NEUTROPENIA	
Grade 1, 2 or Grade 3 (without infection)	Continue cabozantinib at the current dose level. Use of granulocyte colony-stimulating factor support for neutrophil recovery is allowed per investigator discretion and in accordance with accepted guidelines after the first incidence of clinically relevant cytopenia.
Grade 3 with documented infection Grade 4	Interrupt cabozantinib treatment until resolution to grade ≤ 1 , and resume cabozantinib at a reduced dose. Use of granulocyte colony-stimulating factor support for neutrophil recovery is allowed per investigator discretion and in accordance with accepted guidelines after the first incidence of clinically relevant cytopenia.
FEBRILE NEUTROPENIA	
Grade 3	Interrupt cabozantinib treatment until recovery ANC to grade ≤ 1 and temperature to $\leq 38^{\circ}\text{C}$ and resume cabozantinib at a reduced dose
Grade 4	Permanently discontinue study treatment unless determined that the subject is unequivocally deriving clinical benefit. In this case, upon recovery to grade ≤ 1 or baseline, the subject may be re-treated at a reduced dose that is to be determined by the investigator and sponsor and only with approval by the sponsor.
THROMBOCYTOPENIA	
Grade 1, 2 or Grade 3 (without clinically significant bleeding)	Continue cabozantinib at the current dose level.
Grade 3 with clinically significant bleeding or Grade 4	Interrupt cabozantinib treatment until platelet count is $\geq 100\,000/\text{mm}^3$, and resume cabozantinib at a reduced dose
ANEMIA	
Grade 1, 2 or Grade 3	Continue cabozantinib at the current dose level. Supportive care such as red blood cell transfusions may be managed according to institutional guidelines In second occurrence of grade 3, resume cabozantinib at a reduced dose
Grade 4	Interrupt cabozantinib treatment until recovery to grade ≤ 2 and resume cabozantinib at a reduced dose. Supportive care such as red blood cell transfusions should be managed according to institutional guidelines.

For other hematological toxicities (lymphopenia...), the management is at the discretion of the investigator.

8.3.3 Non-hematologic adverse event

Table 2. General approach for the management of cabozantinib-related non-hematologic adverse event	
CTCAE V.5.0	Guidelines/Intervention
Grade 1 and grade 2 AEs which are tolerable	Add supportive care as indicated. Continue cabozantinib at the current dose level.
Grade 2 AEs which are intolerable and cannot be adequately managed	Interrupt cabozantinib treatment or dose reduction. Add supportive care as indicated. If cabozantinib dosing is interrupted, then upon resolution of the AE to baseline or Grade ≤ 1 , or with a dose reduction at the discretion of the investigator unless this is a recurring event at which time the dose should be reduced.
Grade 3 AEs considered related to cabozantinib which occurred without optimal prophylaxis or which is easily managed by medical intervention or resolved quickly	- Interrupt cabozantinib and add supportive care as indicated - For AEs that are easily managed (e.g., correction of electrolytes) with resolution to baseline or Grade ≤ 1 within 24 hours, cabozantinib may be resumed at either the same dose or with a dose reduction at the discretion of the investigator unless this is a recurring event at which time the dose should be reduced - For AEs that require supportive care, the dose should be held while supportive care is initiated and optimized. Then upon resolution of the AE to baseline or Grade ≤ 1, cabozantinib may be resumed at either the same dose or with a dose reduction at the discretion of the investigator unless this is a recurring event at which time the dose should be reduced
Grade 3 AEs considered related to study treatment that occurred despite optimal prophylaxis or is not easily managed by medical intervention	Interrupt study treatment until recovery to $\leq$ Grade 1 or baseline, and resume treatment with a dose reduction
Grade 4 AEs considered related to study treatment	Permanently discontinue study treatment unless determined that the subject is unequivocally deriving clinical benefit. In this case, upon recovery to Grade ≤ 1 or baseline, the subject may be re-treated at a reduced dose that is to be determined by the investigator and sponsor but only with sponsor approval.

8.3.4 Diarrhea

Subjects should be instructed before the beginning of the treatment of the risk of diarrhea and the initial management. A preventive prescription can be given to the patient at the beginning of the treatment.

Subject should notify their physician the first signs of poorly formed or loose stool or an increased frequency of bowel movements. Administration of antidiarrheal agents is recommended at the first sign of diarrhea as initial management. Loperamide is recommended as standard first line therapy. Alternatively, racecadotril can be used. In addition, general supportive measures should be implemented including continuous oral hydration, correction of fluid and electrolyte abnormalities, small frequent meals, and stopping lactose-containing products, high fat meals and alcohol.

Table 3. General approach for the management of cabozantinib-related diarrhea	
CTCAE V.5.0	Guidelines/Intervention
Grade 1 and 2	<i>Administration of antidiarrheal/ antimotility agents is recommended at the first sign of diarrheas initial management.</i> If manageable and tolerable, continue current dose with supportive care. If not tolerable, interrupt dosing and resume at same dose level upon recovery to $\leq$ Grade 1 or baseline. If Grade 2 toxicity recurs, interrupt and restart at next lower dose.
Grade 3	<i>Administration of antidiarrheal/ antimotility agents is recommended at the first sign of diarrheas initial management.</i> Interrupt dosing, unless toxicity can be easily managed with a dose reduction, and resume at the next lower dose level upon recovery to $\leq$ Grade 1 or baseline.
Grade 4	Interrupt dosing immediately. Discontinue study drug unless the patient is deriving clear clinical benefit, as determined by the investigator and the toxicity can be managed with a dose reduction and optimal medical care upon recovery to $\leq$ Grade 1 or baseline.

8.3.5 Nausea and Vomiting

Antiemetic agents are recommended as clinically appropriate at the first sign of nausea and vomiting or as prophylaxis to prevent post-treatment emesis, along with supportive care according to clinical practice guidelines. The 5-HT₃ receptor antagonists are recommended over chronic use of NK-1 receptor antagonists and dexamethasone (NK-1 receptor antagonists can induce or inhibit CYP3A4, and glucocorticoids induce CYP3A4 and thus could lower cabozantinib).

For the management of the cabozantinib, see table 2 non-hematologic adverse event.

8.3.6 Hepatobiliary disorders

Elevations of transaminases have also been observed during treatment with cabozantinib. In general, it is recommended that subjects with elevation of ALT, AST, and/or bilirubin have more frequent laboratory monitoring of these parameters. If possible, hepatotoxic concomitant medications and alcohol should be discontinued in subjects who develop elevated transaminases.

Since subjects may enter the study with elevations of AST/ALT at baseline, the following guideline should be used for dose modifications:

Table 4. General approach for the management of cabozantinib-related Transaminase elevation	
AST and ALT elevation CTCAE V.5.0	Guidelines/Intervention
Subjects with AST and ALT less than or equal to the ULN at baseline	
Grade 1	Continue cabozantinib with weekly monitoring of liver function tests (LFTs) for at least 4 weeks. Then resume the standard protocol-defined monitoring of LFTs.
Grade 2	Continue cabozantinib with at least twice weekly monitoring of LFTs for 2 weeks. Then weekly for 4 weeks. If LFTs continue to rise within Grade 2, interrupt cabozantinib treatment. Then continue with at least weekly LFTs until resolution to Grade ≤ 1. Study treatment may then be resumed at a one-dose-level reduction of cabozantinib
Grade 3	Interrupt cabozantinib treatment and monitor with at least twice weekly LFTs until Grade ≤ 2. Then continue with at least weekly LFTs until resolution to Grade ≤ 1. Cabozantinib may then be resumed at a one-dose-level reduction.
Grade 4	Discontinue study treatment permanently. LFTs should be monitored as clinically indicated, at least 2-3 times per week, until resolution to Grade ≤ 1. If the subject was unequivocally deriving clinical benefit, the subject may be able to resume treatment at a lower dose of cabozantinib as determined by the investigator and sponsor but only with sponsor approval.
Subjects with AST or ALT above the ULN but ≤ 3.0 x ULN (i.e., Grade 1) at baseline	
≥ 1.5 fold increase of AST or ALT AND both AST and ALT are ≤ 5.0 x ULN	Continue cabozantinib treatment with at least twice weekly monitoring of LFTs for 4 weeks and weekly for 4 weeks. If LFTs continue to rise, interrupt study treatment. Then continue with at least weekly LFTs until resolution to Grade ≤ 1. Study treatment may then be resumed at a one-dose-level reduction of cabozantinib
≥ 1.5 fold increase of AST or ALT and at least one of AST or LT is Grade 3 (i.e. AST or ALT > 5.0 but ≤ 20.0 x ULN)	Interrupt study treatment and monitor with at least twice weekly LFTs until Grade ≤ 2. Then continue with at least weekly LFTs until resolution to Grade ≤ 1. Study treatment may then be resumed at a one-dose-level reduction of cabozantinib.
Grade 4	Discontinue study treatment permanently. LFTs should be monitored as clinically indicated, at least 2-3 times per week, until resolution to Grade ≤ 1. If the subject was unequivocally deriving clinical benefit, the subject may be able to resume treatment at a lower dose as determined by the investigator and sponsor but only with sponsor approval.

8.3.7 Thromboembolic events

Events of venous thromboembolism, including pulmonary embolism, and arterial thromboembolism, sometimes fatal, have been observed with cabozantinib. Cabozantinib should be used with caution in patients who are at risk for, or who have a history of, these events. In the HCC study (CELESTIAL), portal vein thrombosis was observed with cabozantinib, including one fatal event. Patients with a history of portal vein invasion appeared to be at higher risk of developing portal vein thrombosis. Cabozantinib should be discontinued in patients who develop an acute myocardial infarction or any other clinically significant thromboembolic complication.

Initiation of therapeutic anticoagulation is allowed on study. During anticoagulation treatment, patients need to be monitored on an ongoing basis for bleeding risk and signs of bleeding which may require additional laboratory tests according to the protocol.

8.3.8 Hypertension

Hypertension has been observed with cabozantinib. Blood pressure should be well-controlled prior to initiating cabozantinib. During treatment with cabozantinib, all patients should be monitored for hypertension and treated as needed with standard anti-hypertensive therapy.

Decisions to decrease or hold the dose of study treatment must be based on blood pressure readings taken by a medical professional and must be confirmed with a second measurement at least 5 minutes following the first measurement.

Table 5. General approach for the management of cabozantinib-related Hypertension	
Criteria For dose Modification	Guidelines/Intervention
> 150 to <160 mm Hg (systolic) OR > 100 to <110 mm Hg (diastolic)	• Optimize antihypertensive treatment by adding new or additional antihypertensive medications and/or increase dose of existing medications. • Reduce cabozantinib treatment by one dose level if optimal antihypertensive therapy (usually to include 3 agents) does not result in BP < 150 mm Hg systolic or < 100 mm Hg diastolic. • If patient is symptomatic interrupt study treatment.
≥ 160 mm Hg (systolic) OR ≥ 110 mm Hg (diastolic)	• Reduce cabozantinib treatment by one dose level and add new or additional antihypertensive medications and/or increase dose of existing medications and monitor patient closely for hypotension. If optimized antihypertensive therapy (usually to include 3 agents) does not result in BP < 150 mm Hg systolic or < 100 mm Hg diastolic, cabozantinib treatment should be dose reduced further or interrupted. • Interrupt cabozantinib treatment if upper limits of BP (≥ 160 mm Hg systolic or ≥ 110 mm Hg diastolic) are sustained and not adequately manageable or if BP is > 180 mm Hg systolic or > 120 mm Hg diastolic or if patient is symptomatic. • Restart cabozantinib treatment at the most tolerable dose and re-escalate cabozantinib dose only if BP falls to and is sustained at < 140 mm Hg systolic and < 90 mm Hg diastolic.
Hypertensive crisis or hypertensive encephalopathy	Discontinue cabozantinib treatment

8.3.9 Corrected QT (QTc) prolongation

Treatment with cabozantinib has been associated with a mild prolongation of the QTc interval. Other factors which may contribute to QTc prolongation include:

- Treatment with other drugs associated with QTc prolongation (see <http://www.qtdrugs.org>)
- Treatment with CYP 3A4 inhibitors (which may increase cabozantinib drug levels)
- Electrolyte changes (hypokalemia, hypocalcemia, hypomagnesemia).
- Medical conditions which can alter electrolyte status e.g., severe or prolonged diarrhea.

If at any time on study there is:

- an increase in QTc interval to an absolute value >500 msec
- or
- an increase of > 60 msec above baseline

➔ two additional ECGs should be performed within 30 minutes after the initial ECG with intervals not less than 3 minutes apart. If the average QTcF from the three ECGs is >500 msec or increased by > 60 msec above baseline, **study treatment must be withheld** and the following actions should be taken:

- Check electrolytes, especially potassium, magnesium and calcium. Correct abnormalities as clinically indicated.
- If possible, discontinue any QTc-prolonging concomitant medications.
- Repeat ECG triplets hourly until the average QTcF is ≤500 msec or otherwise determined by consultation with a cardiologist.

Subjects with QTc prolongation and symptoms must be monitored closely until the QTc elevation has resolved. Cardiology consultation is recommended for evaluation and subject management. Symptomatic subjects must be treated according to standard clinical practice.

Study treatment may be restarted at a reduced dose level if all of the following conditions are met:

- QTc interval to an absolute value < 500 msec and not increase of > 60 msec above baseline
- Symptoms are determined to be unrelated to the QT interval prolongation

Following reinitiation of study treatment, ECGs must be repeated weekly for 2 weeks, then every 2 weeks for 1 month, then according to the protocol-defined timepoints.

Study treatment must be permanently discontinued if either of the following applies:

- Cardiac evaluation confirms that symptoms are the consequence of QT interval prolongation
- Recurrence of QTcF prolongation after reinitiation of study treatment at a reduced dose

8.3.10 Palmar-plantar erythrodysaesthesia syndrome

Palmar-plantar erythrodysaesthesia syndrome (PPES) has been observed with cabozantinib. When PPES is severe, interruption of treatment with cabozantinib should be considered. Cabozantinib should be restarted with a lower dose when PPES has been resolved to grade 1.

Table 6. General approach for the management of cabozantinib-related PPE syndrome	
CTCAE V.5.0	Guidelines/Intervention
Grade 1	Cabozantinib treatment may be continued at the current dose if PPE is clinically insignificant and tolerable. Otherwise, cabozantinib should be reduced to the next lower dose level. Start urea 20% cream twice daily and clobetasol 0.05% cream once daily. If PPE worsens at any time or does not improve after 2 weeks, proceed to the intervention guidelines for Grade 2.
Grade 2	Cabozantinib treatment may be continued if PPE is tolerated. Cabozantinib should be dose reduced or interrupted if PPE is intolerable. Continue urea 20% cream twice daily and clobetasol 0.05% cream once daily and add analgesics (eg, NSAIDs/GABA agonists) for pain control if needed. If PPE does not improve within 2 weeks or worsens or affects self-care, proceed to the intervention guidelines for Grade 3.
Grade 3	Interrupt cabozantinib treatment until severity decreases to Grade 1 or 0. Continue treatment of skin reaction with clobetasol 0.05% cream twice daily and analgesics. Resume cabozantinib at reduced dose if PPE recovers to Grade 1 or 0. Discontinue patient from study if intolerable PPE recurs after dose reduction or does not improve within 6 weeks.

8.3.11 Proteinuria

Proteinuria has been observed with cabozantinib. Urine protein should be monitored regularly during cabozantinib treatment. Cabozantinib should be discontinued in patients who develop nephrotic syndrome.

Table 7. General approach for the management of cabozantinib-related proteinuria	
Criteria For dose Modification	Guidelines/Intervention
≤ 1 g/g (≤ 113.1 mg/mmol)	No change in cabozantinib treatment or monitoring
> 1 to < 3.5 g/g (> 113.1 to < 395.9 mg/mmol)	• No change in cabozantinib treatment required • Confirm with a 24-hour protein assessment within 7 days • Repeat UPCR within 7 days and once per week. If UPCR < 1 on 2 consecutive readings, UPCR monitoring can revert to protocol-specific times. (Second reading is confirmatory and can be done within 1 week of first reading.). Ask nephrology medical advice to optimize nephroprotective treatment.
≥ 3.5 g/g (≥ 395.9 mg/mmol)	• Hold cabozantinib treatment and ask nephrology medical advice and repeat UPCR within 7 days and/or 24-hour urine protein. • If ≥ 3.5 on repeat UPCR, continue to hold cabozantinib treatment and check UPCR every 7 days. If UPCR decreases to < 2, restart cabozantinib treatment at a reduced dose and monitoring of urine protein/creatinine should continue weekly until the UPCR decreases to < 1.
Nephrotic syndrome	Discontinue cabozantinib treatment

8.3.12 Reversible posterior leukoencephalopathy syndrome

Reversible Posterior Leukoencephalopathy Syndrome (RPLS), also known as Posterior Reversible Encephalopathy Syndrome (PRES), has been observed with cabozantinib. This syndrome should be considered in any patient presenting with multiple symptoms, including seizures, headache, visual disturbances, confusion or altered mental function. Cabozantinib treatment should be discontinued in patients with RPLS.

8.3.13 Osteonecrosis

Cases of osteonecrosis of the jaw have been observed with cabozantinib. Oral exam should be performed before initiation of cabozantinib and regularly during treatment with cabozantinib. The practice of oral hygiene rules should be recommended to patients.

Cabozantinib treatment should be discontinued in patients who develop osteonecrosis.

In case of dental surgery or invasive dental treatment, cabozantinib should be suspended for at least 28 days before surgery. Therefore, the patient should be withdrawn of the study.

9 BIOLOGICAL ANCILLARY STUDY

9.1 TUMOURS

Possible approaches to be developed with additional grants may include for example: immune-monitoring of immune infiltrate, evaluation of PD-L1 expression on tumour and stromal cells by IHC, evaluation of CTLA4, CD8, OX40L and TIGIT expression by IHC, evaluation of mRNA expression of CD8, OX40L, TIGIT, CTLA4, PD1 and PD-L1 by real-time quantitative RT-PCR, molecular tumour characterization using NGS-array-CGH-transcriptomics and whole-exome or whole genome for selected samples and IHC evaluation of MET.

These analyses will be performed on:

- In optional: Fresh biopsy (one paraffin block) will be obtained at baseline from primitive and/or from metastatic sites if feasible and if the patient agrees on the consent form.

AND

- Mandatory: Archival initial or recurrent tumour sample
One paraffin block will be sent to pathologist sponsor during study. This sample will be returned to centre after analysis.

9.2 BLOOD SAMPLES

Blood samples will be taken at baseline, W4 (D1C2), W6 (D15C2, first radiological assessment) and at progression.

Translational further researches may be used to:

- Perform NGS analysis of constitutional DNA
- Identify HPVctDNA and to follow it up during treatment.
- Identify potential plasma biomarkers of cabozantinib by measuring a panel of circulating pro-angiogenic and pro-inflammatory biomarkers: VEGF, placental growth factor (PIGF), soluble (s) VEGFR1/FLT-1, basic fibroblast growth factor (bFGF), VEGF-C, VEGF-D, TIE-2, interleukin (IL)-1 β , IL-2, IL-4, IL-6, IL-8, IL-10, IL-12p70, interferon (IFN)- γ , and tumor necrosis factor (TNF)- α , HGF, sVEGFR2, angiopoietin-1 (Ang-1), Ang-2, tissue inhibitor on matrix metalloproteinase (TIMP)-1, carbonic anhydrase (CA)-IX, free insulin-like growth factor (IGF)-I and stromal-derived factor (SDF1)- α , PDL1, AXL and Met.

9.3 NUTRITIONAL ANCILLARY STUDY

9.3.1 Anthropometry and Body Composition

Regional muscle will be measured at baseline and during treatment by using CT images. Two consecutive slices from third lumbar vertebrae (L3) landmark extending to the iliac crest will be obtained for each 25. Image sections will be analyzed by a trained person using SliceOmatic software V5.0 (Tomovision) and established thresholds of skeletal muscle tissue density (-29 to +150 Hounsfield units) (Mitsiopoulos et al, 1998).

Directly measured body composition variables will be total lumbar skeletal muscle cross-sectional area (cm²) and total lumbar adipose tissue cross-sectional area (cm²). Cross-sectional areas of the sum of all L3-containing muscles (psoas, paraspinal and abdominal wall muscles) will be measured in cm² unit for each slice and the average value of both consecutive images will be used as total muscle area (TMA) for each patient. TMA will be normalized for stature to calculate skeletal muscle index (SMI), expressed in cm²/m². Sarcopenia will be defined as SMI lower than a pre-specified sex-based threshold: 55.4 for men and 38.9 for women.

9.3.2 radiology review

A central review of the scanners will be done in the study to assess sarcopenia. An anonymized copy of the scanner imagery will be sent to the sponsor during the study.

10 EFFICACY AND SAFETY CRITERIA

10.1 EFFICACY CRITERIA

Response to treatment will be evaluated according to RECIST 1.1 criteria.

10.2 SAFETY CRITERIA

The safety assessment will be done by evaluating the general condition (PS, weight) and clinical status of the patients, and by collecting the events occurring during and after the treatment during the visits.

A special attention will concern fistula/perforation adverse event.

The intensity of the events will be estimated according to the NCI-CTCAE version 5.0 classification (grade 1 to 5 toxicity).

The intensity of adverse events not listed in this classification will be graded as follows:

- Grade 1: mild Discomfort noticed but no disruption of normal daily activity
- Grade 2: moderate Discomfort sufficient to reduce or affect daily activity; no treatment or medical intervention is indicated although this could improve the overall well-being or symptoms of the patient
- Grade 3: severe Inability to work or perform normal daily activity; treatment or medical intervention is indicated in order to improve the overall well-being or symptoms; delaying the onset of treatment is not putting the survival of the patient at direct risk
- Grade 4: Life-threatening/disabling
An immediate threat to life or leading to a permanent mental or physical conditions that prevents work or performing normal daily activities; treatment or medical intervention is required in order to maintain survival
- Grade 5: Death AE resulting in death

11 SAFETY

11.1 GENERAL RULES – INSTRUCTIONS

Safety management will be conducted according to the French regulatory requirements (Law No. 2012-300 of 05/03/2012 as amended by Ordinance No. 2016-800 of 16 June 2016)

11.2 DEFINITIONS

11.2.1 Adverse Event (AE)

Any untoward medical occurrence in a patient or clinical investigation subject administered a pharmaceutical product and which does not necessarily have a causal relationship with this treatment. An adverse event (AE) can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the use of a medicinal (investigational) product, whether or not considered related to it.

11.2.2 Adverse Reaction

All untoward medical occurrences in a patient or clinical investigation subject who participate to research involving the human person **related to the study procedure or to the experimental product**.

All untoward responses to an investigational medicinal product related to any dose administered.

11.2.3 SERIOUS Adverse event

A serious adverse event (SAE) is an AE that fulfils one or more of the following criteria:

- Results in death
- Is immediately life-threatening
- Requires in-patient hospitalization or prolongation of existing hospitalization
- Results in persistent or significant disability or incapacity
- Is a congenital abnormality or birth defect
- Is an important medical event*.

* Medical and scientific judgment should be exercised in deciding whether expedited reporting is appropriate in other situations, such as important medical events that may not be immediately life-threatening or result in death or hospitalization but may jeopardize the patient or may require intervention to prevent one of the other outcomes listed in the definition above. These should also usually be considered serious.

Some “hospitalization/prolonged hospitalization” are not considered to be reported as serious adverse event:

- Admission for administrative or social reasons
- Hospitalization predefined by the protocol
- Hospitalization for medical or surgical treatment planned before the beginning of the trial
- Passage to day hospital
- Hospitalizations for pre-existing signs and symptoms that have not been aggravated

11.2.4 Suspected Unexpected Adverse Reaction

Any serious adverse reaction for which the nature, severity, frequency or outcome is not consistent with the applicable product information (e.g investigator’s brochure for an unauthorized investigational product or summary of product characteristics for an authorised product).

Any serious adverse reaction for which the nature, severity, frequency or outcome is not consistent with the applicable product information, diagnostic or therapeutic study mandated procedure (non-invasive and/or invasive) evaluated in the protocol.

11.3 INVESTIGATOR’S RESPONSABILITIES

11.3.1 Detection and registration of adverse events

All adverse events have to be searched, reported and recorded, processed and evaluated.

All adverse events occurring, once the informed consent form has been signed, **during treatment and 30 days after the last study drug administration** must be recorded by the investigator.

The intensity of the events will be estimated according to the NCI-CTCAE version 5.0 classification (grade 1 to 5 toxicity). The intensity of adverse events not listed in this classification will be graded as follows:

- **Grade 1: mild** : Discomfort noticed but no disruption of normal daily activity
- **Grade 2: moderate** : Discomfort sufficient to reduce or affect daily activity; no treatment or medical intervention is indicated although this could improve the overall well-being or symptoms of the patient
- **Grade 3: severe** : Inability to work or perform normal daily activity; treatment or medical intervention is indicated in order to improve the overall well-being or symptoms; delaying the onset of treatment is not putting the survival of the patient at direct risk
- **Grade 4: Life-threatening/disabling** : An immediate threat to life or leading to a permanent mental or physical conditions that prevents work or performing normal daily activities; treatment or medical intervention is required in order to maintain survival
- **Grade 5: Death AE or resulting in death**

The investigator will specify for each event:

- ✓ The seriousness
- ✓ The date of start +/- the date of end of the event
- ✓ The duration
- ✓ The intensity
- ✓ The causality assessment
- ✓ The outcome

If a SAE occurs, the sponsor should be notified of the event awareness by the investigator without any delay. This will be done by mailing sponsor's Serious Adverse Event Report Form:

Adverse Events based on examinations and tests

The reporting of laboratory/vital signs/ECG abnormality as AE should be avoided unless one of the following is met:

- Any criterion for an SAE is fulfilled
- Causes study treatment discontinuation
- Causes study treatment interruption
- Causes study treatment dose reduction
- The investigator believes that the abnormality should be reported as an AE

If deterioration in a laboratory value/vital sign is associated with clinical signs and symptoms, the sign or symptom will be reported as an AE and the associated laboratory result/vital sign will be considered as additional information. Wherever possible the reporting investigator uses the clinical, rather than the laboratory term (eg, anemia versus low hemoglobin value). In the absence of clinical signs or symptoms, clinically relevant deteriorations in non-mandated parameters should be reported as AE(s).

11.3.2 SAE reporting process

The investigator has to report the Sponsor of all serious adverse events occurring, once the informed consent form has been signed, **during treatment and 30 days after the last study drug administration.**

All late Serious Adverse Events and considered as reasonably related to the study treatment(s) or to the study procedure must be reported without delay limitation.

The investigator has to immediately report to the sponsor all serious adverse events with the exception of those that are identified as not requiring immediate reporting in the protocol

This will be done by mailing sponsor's Serious Adverse Event Report Form, and it will be following by some report, as needed, containing all available information concerning the SAE.

Pharmacovigilance Unit: pv-cfb@baclesse.unicancer.fr

The investigator will specify for each event:

- ✓ The description as clear as possible according to the medical terminology,
- ✓ The intensity,
- ✓ The date of start and the date of end of the event,
- ✓ Measures taken and the necessity or not for a corrective treatment,
- ✓ If the treatment of the study was discontinued or if dose was modified
- ✓ The outcome: For a non-fatal event, the event should be followed until resolution or return to the initial status or to stabilization of potential sequels,
- ✓ The **causality assessment**: he has to assign a causality of "related to study treatment", if there is a "reasonable possibility" that the study treatment caused the event, or "not related to study treatment" if there is "no reasonable possibility" that the study treatment caused the event. The investigator must also assess whether the serious adverse event is "possibly related" to any study mandated procedure or activity, or others concomitants treatments, or under study or any others disease.

The investigator should also, when possible, attach to the serious adverse event report:

- ✓ A copy of the hospital report/hospitalization prolongation,
- ✓ A copy of the autopsy report,
- ✓ A copy of the results of all additional exams performed, including relevant negative results together with the range of normal laboratory values,
- ✓ Any other document judged useful and pertinent (including imaging)

All these documents should be **anonymized**.

Additional information may be requested by the sponsor.

The investigator should document the event as best as possible, provide medical diagnosis where possible and establish a causal link between the serious adverse event and the research, the study treatment, the associated drugs, an underlying pathology, progression of the disease or other cause.

The investigator shall promptly provide the sponsor with additional information regarding serious adverse events as he becomes aware of them.

The investigator should follow the patient who has had a serious adverse event until resolution, stabilization at a level acceptable to the investigator or return to the previous status, even if the patient withdraws from study and inform the sponsor of the serious adverse event outcome.

11.3.3 Special cases

ADVERSE EVENT OF SPECIAL INTEREST

An adverse event of special interest (AESI) is a serious or non-serious adverse event that requires special attention and will be specifically searched. AESIs should be notified and followed as serious adverse events.

As part of this research, the following events are considered as AESI:

-FISTULA:

- › Gastrointestinal fistula: anal, colonic, ileal, rectal,...
- › Urinaryfistula,
- › Genital tract fistula: uterine, vaginal...

-GASTROINTESTINAL PERFORATION

REPORTING EXCEPTIONS

Serious adverse event not to be notified immediately

Any event that is part of the natural history of the disease (progression of the disease or hospitalization for progression of the disease) should not be notified to pharmacovigilance on the SAE form but has to be reported into the e-CRF.

Some "hospitalization/prolongation of hospitalization" are not considered as serious adverse events and do not require notification to pharmacovigilance (see last paragraph "Serious adverse event").

PREGNANCY

If a woman starts a pregnancy as part of the study or in some cases if her partner participates in the study (drug that can reach the seminal line of the man), the investigator has to report pregnancy to the sponsor.

The investigator informs the sponsor who will send him the "pregnancy notification form". The investigator has to follow the patient until the end of the pregnancy or its interruption and notify the outcome to the sponsor. If the outcome of pregnancy falls within the definition of serious adverse events (spontaneous abortion with hospitalization, fetal death, congenital anomaly, ...) the investigator should follow the SAE reporting process.

12 STATISTICAL CONSIDERATIONS

12.1 NUMBER OF SUBJECTS

The total sample size to be enrolled is **57** patients.

As it is the first time the cabozantinib is used for advanced cervical carcinoma after platinum treatment failure, we plan to use a single arm two-stage **multicentre phase II based on a Bryant-and-Day design**. This single arm two-stage design was selected in order to simultaneously assess efficacy and safety and minimize the expected number of patients treated in the event that cabozantinib monotherapy proves to be very disappointing (in terms of efficacy and/or safety) in this setting.

According to literature, median PFS for advanced cervical cancer patients is around 3.4 months under bevacizumab monotherapy (Monk et al.2009) and 4 months under pazopanib treatment (Monk et al. 2010). As for tolerance profile, fistula formation was noticed for around one quarter of advanced cervical cancer patients under bevacizumab (Godoy-Ortiz et al., 2018; Palavalli Parsons et al., 2017). By extrapolation, we expect for:

Efficacy: $\pi_{\text{Efficacy}0} = 30\%$, $\pi_{\text{Efficacy}1} = 50\%$, the unacceptable and acceptable 3-month control rates

Toxicity: $\pi_{\text{Toxicity}0} = 25\%$, $\pi_{\text{Toxicity}1} = 10\%$, the unacceptable and acceptable fistula rates

Assuming the error rates:

-One-sided $\alpha_{\text{Efficacy}}=5\%$ and $\alpha_{\text{Toxicity}}=10\%$

-Power=80%

Stage 1: 22 assessable patients will be included for interim analysis.

Initially, Enrolment should not be halted to conduct the interim analysis.

However, due to the quickness of recruitment in the study, the protocol is amended to suspend the inclusions during the interim analysis, in order to have a sufficient perspective on the tolerance and efficacy of the treatment.

Then:

- ⇒ if there are $p_{\text{Efficacy}} < 8/22$ **OR** $p_{\text{Toxicity}} > 4/22$ the study will STOP.
- ⇒ Otherwise the second stage (stage 2) will be planned with 29 additional assessable patients (for a total of N=51 patients).

At the final analysis:

If $p_{\text{Efficacy}} < 21/51$ **OR** $p_{\text{Toxicity}} > 9/51$: the study will be considered as negative.

Assuming a 10% drop-out rate, a **total of 57** (25 for the stage 1 and 32 additional for the stage 2) patients will be included in the present study.

12.2 PATIENTS UNDER BEVACIZUMAB

A minimum of 19 evaluable bevacizumab pre-treated patients among the total 57 recruited patients is required. Using a 1-stage A'hern design (Reference: A'Hern RP. Sample size tables for exact single-stage phase II designs. Stat Med. 2001 Mar 30;20(6):859-66.) in the bevacizumab pre-treated cohort, would allow to accept 3-month control rates $\geq 50\%$ while rejecting a rate of $\leq 20\%$ with 80% power at a 2.5% significance level. The resulting decision rule requires to observe **≥ 8 "alive-controlled"** (responding to the treatment) patients at 3 months out of 19 evaluable patients in the bevacizumab pre-treated cohort to reject the null hypothesis ($\leq 20\%$) in favor of the alternative ($\geq 50\%$).

If the limit of 19 patients is not reached, efficacy and safety of the bevacizumab subgroup will be described in detail and we will further compute the reached statistical power.

Feasibility

The estimated accrual for this study is 3 patients a month. Thus, patient accrual for the 57 patients is expected to be completed within 18 months. Additional 3 months are required to allow the response data to mature.

12.3 STATISTICAL ANALYSIS

Patients will be evaluable for the efficacy analysis if they have a reported progression ≤ 3 months or a minimum follow-up of 3 months. Patients who drop out from the study prior to the 3 months will be considered to have failed treatment and therefore will not be included in the efficacy analysis but they will be included as failed treatment in the intent-to-treat analysis.

Patients will be included in the safety analysis if they received at least 1 dose of cabozantinib.

Patients who were removed from the study due to adverse events will be followed-up until resolution or stabilization of symptoms.

Efficacy and safety will be evaluated simultaneously for main objective.

The primary endpoints, the response rate and toxicity rate, will be evaluated at 3 months with their corresponding two-sided 95% CI.

For time-to-event endpoints, medians (if reached) will be presented and/or event rates at selected time points using the Kaplan-Meier method.

Frequency tables will be tabulated (overall and by tumor group) for all categorical variables by the levels of the variables and proportions will be estimated using the efficacy population as denominator (unless specifically specified otherwise). All estimates will be complemented with an appropriate 95% confidence interval where applicable. Sensitivity analyses will explore efficacy and safety between patients previously treated or not with cabozantinib.

Adverse events and laboratory abnormalities observed during the study will be tabulated (worst CTC grade per patient) overall and further tabulation will be made based on time of occurrence and relationship to treatment. The latter excludes events unrelated or not likely related to treatment, but includes events for which the relationship with treatment is not assessable. Tolerability will be summarized by duration of treatment, reasons of discontinuation, dose reduction rates and reasons for dose reductions.

Patient reported outcomes

Quality of life (QoL) scores and changes from baseline scores will be described for selected primary scales. Missing values will be calculated such that if at least half the items from the scale will be completed, it will be assumed that the missing items will have values equal to the average of those items present. The Z-test or the non-parametric Wilcoxon–Mann–Whitney tests will be used to evaluate the evolution in global health status and other dimensions of the EORTC QLQ-C30 and subscale QLQ-CX24. In addition, an evaluation of the time to ≥ 10 points deterioration (TTD) will be also carried out using survival analysis methods. A composite definition for deterioration based on death, tumor progression will be used. To account for missing data as a potential major source of bias, the compliance mechanism will be investigated prior to initiating the QoL analysis. Characteristics of patients with and without valid QoL data will be compared and trends over time per dropout pattern will be investigated. Proportional hazard Cox regression model will be used in order to investigate whether the compliance mechanism is linked to selected prognostic variables.

13 . QUALITY CONTROL

13.1 INDEPENDENT DATA MONITORING COMMITTEE (IDMC)

An Independent Data Monitoring Committee (IDMC) will be set-up to ensure the protection of patients, to ensure the ethical conduct of the study, to evaluate the benefit/risk ratio of the study and to insure an independent review of the scientific outcomes during and at completion of the study. The Independent Committee exercises a consultative role for the promoter who takes the final decision for implementing the recommendations proposed by the committee.

The committee will include a statistician, a pharmacologist and a clinical physician.

This committee will be consulted:

- **Before the trial initiation**
- **After inclusion of 10 patients for a safety run. A special attention will concern fistula/perforation adverse event.**
- **At interim analysis**
- **During at the final analysis**

Dealing with safety, the committee will be informed by the Sponsor of all the observed toxicities (SAEs) for regular reviewing of safety data and advice. It should propose to stop inclusions in case of unacceptable toxicities (in terms of severity and/or frequency).

For this purpose, regular teleconferences will be planned and will contain two parts: the first one with the Sponsor introducing the teleconference with a safety review, and the second one with only IDMC members who will independently deliberate. The Sponsor will be in charge of reporting all safety concerns to the IDMC, in terms of grade and timing along with treatment protocol adherence.

All the decisions will be documented in writing and communicated to all the investigators.

If needed, ad hoc teleconferences may be organized for any safety alert.

13.2 QUALITY ASSURANCE

In order to ensure the authenticity and credibility data in accordance to Good Clinical Practices (GCP), the sponsor will implement a quality assurance system that includes:

- the management of the test according to the procedures of the Clinical Research Unit
- the quality control of the data of the investigator site by the monitor whose role is to check the concordance and the coherence of the data of case report form of observation compared to the documents-source
- the provision if the funding provides for dedicated staff in the service to assist the investigator in the logistics of the study and the collection of data in the case report form.

14 ETHICS AND REGULATORY CONSIDERATIONS

The study will be conducted in accordance with the French Public Health Law, specially relating to research involving the biomedical human person of the Public Health Code, articles L1121-1 and following (Law No. 2012-300 of 05/03/2012 as amended by Ordinance No. 2016-800 of 16 June 2016), the Bioethics Law, the law related to the protection of physical persons for the treatments of personal data and related to information technology, database and liberties, the Helsinki declaration and the Good Clinical Practices

14.1 CLINICAL TRIAL AUTHORISATION

An authorization request will be sent by the sponsor to the French regulatory authorities before the study initiation:

- Ethics Committee (Committee for the Protection of Persons, CPP)
- Competent Health Authority (ANSM)

An Information will be given to the Competent Authority (ANSM), with transmission of the synopsis of the study and the favorable opinion of the CPP.

This study is under of the "Reference Methodology" (MR-001) in application of the provisions of article 54 paragraph 5 of the law of 6 January 1978 as amended relating to data processing, files and freedoms. This change was approved by decision of 5 January 2006.

The François Baclesse Center respects the regulations in force, in particular the rights of the persons being treated according to the EU regulation 2016/679 on the protection of data ("RGPD").

Any substantial modification in the protocol about objectives, design, population, evaluation, significant administrative modifications will need the coordinator approval, the sponsor approval, CPP approval and the competent authority authorization.

14.2 INFORMATION OF PATIENTS INVOLVED IN THE RESEARCH

Patients will be completely and faithfully informed with understandable words on the objectives and constraints of the research, potential risks, required measures for monitoring and safety, of their right to decline the participation in the study or the possibility to withdraw from the study at any time. All these informations are included in the informed consents form given to the patient: one for the main study, other for ancillary studies. The investigator, or the physician who represents him, will collect the signed written informed consent(s) before the definitive inclusion in the study. A copy of the information and consent form signed by the two parties will be given to the patient; the investigator will keep the second copy.

For any significant modification of the protocol related to the objectives of the research, its design, the population, the exams or significant administrative aspects, a new consent from each person participating to the research will be collected if needed.

14.3 INVESTIGATOR RESPONSIBILITIES

The Principal Investigator of each participating center is committed to conduct the clinical trial in accordance with the trial protocol and with the regulations in force, notably the decision of 24 November 2006 related to Good Clinical Practices.

The Principal Investigator is responsible for:

- ✓ Giving to Sponsor his/her curriculum vitae and that of co-investigators
- ✓ Identifying persons involved in the research in his/her team and defining their responsibilities
- ✓ Initiating the inclusion of patients after Sponsor authorization
- ✓ Making the maximum effort to include the required number of patients within the established recruitment period.

Each Investigator is responsible for:

- Obtaining the signed and dated informed consent and personally signing this consent for each participating patient before any procedure specific to the trial
- Regularly completing the Case Report Form (CRF) for each patient included in the trial and to allow to CRAs mandated by the Sponsor a direct access to source data in order to validate the data entered in the CRF
- Dating, correcting and signing any correction in the CRFs and data clarification forms (DCF)
- Accepting the regular monitoring visits of the monitor and eventual auditors mandated by the Sponsor or inspectors of supervisory authorities

Source documents, defined as any document or original item that allow to prove the existence or accuracy of a data or a fact recorded during the study, will be kept during 15 years by the investigator or the hospital if the source is a hospital medical record.

The archiving of the data will be the responsibility of the investigator and according to the legislation. The patient should keep the data and a patient identification list for a minimum of 15 years after the end of the study.

14.4 DATA CONFIDENTIALITY

The investigator will ensure the confidentiality of all information concerning the project for himself and for all persons involved in the conduct of the trial until the publication of the test results. This confidentiality obligation will not apply to information that the investigator will be required to provide to patients in the context of their participation in the trial or to information already published. The investigator will ensure not to publish, disclose or use, in any way, directly or indirectly, scientific or technical informations of the trial.

The study may not be the subject of any written or oral commentary without the agreement of the sponsor; all the information communicated or obtained during the realization of the test belonging in full right to the sponsor who can freely dispose of it.

15 DATA AND DOCUMENTS KEEPING

15.1 DATA ENTRY AND HANDLING

Data management will be performed by the Data Processing Center (CTD) of the North West Cancéropôle (Centre de Traitement des Données du Cancéropôle Nord-Ouest). The CTD provides a database management software dedicated to clinical research: Ennov Clinical (version 7.5.10, ENNOV / CLINSIGHT, 33155 Cenon, France).

This software package, which is based on an Oracle database architecture, is designed for the overall management of clinical and epidemiological studies, meets the regulatory requirements related to this type of study. The CTD Ennov Clinical instance is validated in its computing environment. A data validation plan will be developed jointly by the Clinical Research Unit and the Data Processing Center and will describe in detail the controls to be performed for each variable.

A database specific to the study will be created, tested and validated before the start of the study. All information required by the protocol must be recorded on the paper observation books - or on the electronic observation booklet - under the responsibility of the principal investigator and an explanation must be provided for each missing data item. The data will have to be entered in these notebooks as they are obtained, and the sponsor will take over the monitoring. The data will then be checked by the CTD in accordance with the data validation plan. The database will be frozen after final quality control and then exported to the adequate format for statistical analysis according to an automated and validated procedure.

15.2 ARCHIVING

The sponsor must ensure the archiving of essential documents on the conduct of the study in conditions ensuring their safety, for the minimum duration provided by BPC, 15 years after the end of the research. These documents are: the protocol and annexes including any amendments, original signed information forms and consents, questionnaires, case report forms, follow-up documents, statistical analysis, the final report of the study.

15.3 PUBLICATION POLICY

The results of this study, property of the Sponsor (Centre François Baclesse), will be published under scientific articles. Publications relating to or resulting from this research will be communicated and submitted for review by the study coordinators to all investigators.

The authors include investigators that have included most patients, the biostatistician who has performed the data analysis, the clinical researcher monitor and the participants who provided substantial contribution to the development of the study, the analysis and interpretation of results and / or the writing of the manuscript.

Acknowledgement to IDMC members will be indicated in final publication.

No publication or presentation of the results will be allowed without the agreement of all the parties. Each investigator will be author in the order determined by the number of eligible patients included. No publication or communication will be performed without the agreement of the coordinating investigator and the Sponsor with the obligation to mention the name of the Sponsor, the support of Ipsen Pharma SAS which financially supported the conduct of the trial and provided Cabozantinib.

For further ancillary studies on the biological collections, some dedicated publications will be also performed.

This work will be the property of all authors and will be at their disposal for transversal communications and publications. Publications related to the results of potential ancillary studies need prior approval of the coordinating investigator and methodologist and will be done after the publication of the main study, which should be cited as reference.

16 FUNDING AND INSURANCE

16.1 FUNDING

Any additional costs referred to the Code of Public Health are being negotiated between the CFB and the representative of the institution, taking into account the financial resources available to the CFB in the frame of its public promotion activities.

However, the CFB will ensure the study implementation and supply of the following material (protocol, CRF, investigator file) needed to the conduct of the study. In the case of equipment or treatments are provided by other partners, the conditions must be specified in the study agreement.

16.2 INSURANCE

The sponsor has subscribed for the duration of the study an insurance covering his own liability and that of any physician involved in the realization of the study. It will also ensure full compensation for the harmful consequences to search for the person undergoing it and assigns, unless evidence against him that the damage is not attributable to its fault or that of any intervener, without that can be opposite the act of a third party or the voluntary withdrawal of the person who had originally agreed to participating to research. (Article L 1121-10)

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18 ANNEXE - Fridericia formula

The Fridericia formula is depicted below for calculation of QTcF.

$$QTcF = \frac{QT}{RR^{1/3}}$$

[QT = measured QT interval in milliseconds; RR = measured R to R interval (which can be derived from the heart rate [HR] as 60/HR)]